VID – Complete Workspace List

S.No	Workspace	Key Modules / What It Contains
1	Academic Admin / Coordinator	Digital Noticeboard, Activities, Principal Dashboard, Academics (Classes, Batches, Details, Exam Dates), Academic Setup (Curriculum, Holidays, Calendar, Timetable Slots, Sections), Courses, Master Courses, Assessments (Types, Templates), Data Management (Import/Export), Configuration (OBE, Question Papers, LTI), What's New
2	Admission Counselor	Applications, Admissions, Configuration, Enquiry Handling, What's New
3	Admission Officer	Prospects, Applications, Admissions Workflow, Student Onboarding
4	Academic Coordinator (Operational)	Digital Noticeboard, Course Registration, Student Promotions, All Students, Question Banks, Courses, Master Courses, Attendance, Classrooms, Reports, Faculty Timetable
5	Student Workspace	Profile, Attendance, Timetable, Exams, Results, Fees, LMS, Activities
6	Faculty Workspace	Classes, Attendance, Assignments, Question Banks, LMS, Student Performance, Communication
7	Examination Workspace	Exam Setup, Scheduling, Marks Entry, Report Cards, Results, Analytics
8	Payment Workspace	Academic Fees, Exam Fees, Reports, Concessions, Online Transactions, Challans, Settlements, Scholarships, Credit Memos, Fee Cards, Permissions, Restrictions, Open Payments, Configuration
9	HRMS / Employee Workspace	Home, Attendance, Leaves, Salary, Expenses, Shift Requests, Compensatory Leave
10	Timetable Workspace	Timetable Creation, Slot Management, Faculty Allocation, Substitutions
11	Common Workspace	Home, Student Central, Service Requests, My Requests, Payments, Exams, HRMS, Configuration, What's New
12	User Management System (UMS)	User Creation, Role Assignment, Permissions (RBAC), Identity Management, Access Control
13	Principal / Admin Dashboard	Analytics, Reports, KPIs, Performance Monitoring, Insights

14	Team Owner (Super Admin)	Institutions, Organization Settings, User Management, Apps & Integrations, Data Management, Branding (Logo, Favicon), Reports
15	Transport Coordinator	Digital Noticeboard, Transport Analysis, Registration, Buses, Boarding Points, Routes, Route-Bus Mapping, Assignments, Configuration, What’s New
16	Library Management	Book Inventory, Issue/Return, Tracking, Usage Analytics
17	Parent Portal	Student Progress, Attendance, Fee Payments, Communication
18	Event Management	Event Planning, Scheduling, Participation Tracking
19	Disciplinary Committee	Incident Reports, Actions, Behavior Tracking
20	Sports Officer	Events, Teams, Performance Tracking, Achievements
21	Health Monitoring	Medical Records, Emergency Logs, Alerts
22	Hostel Administration	Dashboard, Hostellers, Attendance, Gate Pass Requests, Reports, Configuration, What’s New
23	Alumni Management	Alumni Database, Engagement, Communication
24	Placement Cell	Job Drives, Applications, Company Interaction, Career Tracking
25	Research & Development	Projects, Publications, Innovation Tracking
26	Course Coordinator	Digital Noticeboard, All Courses, Curriculum Mapping, What’s New
27	YANTRA (AI Engine)	Chatbot, Voice Agent, Face Recognition Attendance, AI Automation, Smart Support

WORKSPACE 1: ACADEMIC ADMIN / COORDINATOR WORKSPACE

1. 🧭 Overview

The **Academic Admin Workspace** is the **core brain of the institution's academic system**. It is responsible for defining, configuring, and controlling the entire academic structure of the school.

This workspace is not used for daily teaching operations — instead, it is used to **design the academic ecosystem** that all other workspaces depend on.

It acts as:

- The **foundation layer** for all academic data
- The **configuration engine** for curriculum, classes, and policies
- The **control panel** for academic planning and structure

2. 👥 Primary Users

- Academic Coordinator
- Principal / Vice Principal
- Institution Admin

These users have **high-level permissions** and can modify system-wide academic configurations.

3. 🧩 Complete Module Breakdown

3.1 Digital Noticeboard & Activities

This module is used to broadcast academic-related updates.

Capabilities:

- Publish academic notices (exam schedules, curriculum updates)
- Announce academic activities (competitions, assessments)
- Target specific classes/sections
- Track acknowledgment (who viewed/read)

3.2 Academics (Core Structure Management)

This is the **most critical module**.

What it manages:

- Classes (Grade 1, Grade 2, etc.)
- Batches (Academic Year like 2025–26)
- Sections (A, B, C divisions)
- Academic hierarchy

Detailed Features:

- Create new academic programs (School-level or Grade-level)
- Define batch lifecycle (start date, end date)
- Configure class structure (e.g., Grade 1 → Sections A, B)
- Store metadata (capacity, class teacher, room info)

Editable Actions:

- Modify class details
- Reassign sections
- Update academic timelines

3.3 Academic Setup

This module defines **rules and structure of academics**.

Includes:

a) Curriculum Configuration

- Subject list per class
- Subject types (Core / Elective / Lab)
- Subject credits (optional for advanced systems)

b) Academic Calendar

- Term structure (Term 1, Term 2)
- Holidays (national, school-specific)
- Exam windows
- Events and activities

c) Timetable Slots

- Define period structure (Period 1, Period 2…)
- Duration per period
- Break slots (Lunch, Short Break)

d) Sections Management

- Create and manage class divisions
- Assign capacity limits
- Map students later

3.4 Courses & Course Types

This module defines **what is being taught**.

Features:

- Create course categories (Academic / Co-curricular)
- Define subject-level courses (Math, Science, English)
- Assign course types (Theory / Practical)

3.5 Master Courses

Acts as a **central repository of all courses**.

Purpose:

Avoid duplication and maintain standardization.

Features:

- Global course definitions
- Reusable course templates
- Version control (if curriculum changes)

3.6 Assessments & Evaluation System

This module defines **how students are evaluated**.

Includes:

a) Assessment Types

- Unit Test
- Mid-Term
- Final Exam
- Assignment-based evaluation

b) Assessment Templates

- Marks structure (e.g., 50 marks theory + 50 practical)
- Grading system (A, B, C or percentage)

c) Evaluation Rules

- Pass criteria
- Weightage rules
- Aggregation logic

3.7 Data Management

Handles **bulk data operations**.

Capabilities:

- Import student data (CSV/Excel)
- Export academic data (reports, structures)
- Bulk update records

Use Cases:

- Migrating data from old systems
- Backups and audits

3.8 Configuration (Advanced Settings)

This is a **technical + academic configuration layer**.

Includes:

a) Question Paper Templates

- Define exam paper structure
- Section-wise marks distribution

b) OBE (Outcome-Based Education) Configuration

- Define learning outcomes
- Map subjects to outcomes
- Track achievement levels

c) LTI Configuration

- Integration with external learning tools (LMS platforms)

d) System Rules

- Promotion criteria
- Attendance rules
- Academic policies

3.9 "What's New" Module

Purpose:
Keeps users updated with:

- New features
- System updates
- Academic changes

4. 🔄 End-to-End Workflow

1. Create Academic Year
2. Define Classes & Sections
3. Configure Curriculum & Subjects
4. Set Academic Calendar
5. Define Timetable Slots
6. Setup Assessment Rules
7. Publish Academic Structure

This workflow ensures all downstream modules (student, faculty, exams) work correctly.

5. 🧠 Data Ownership

This workspace owns:

- Academic Year Data
- Class & Section Structure
- Subject & Course Definitions
- Assessment Rules
- Academic Policies

Other workspaces **consume this data** but do not modify it.

6. 🔗 Dependencies

Used by:

- Student Workspace (for class & timetable)
- Faculty Workspace (for subjects & teaching)

- Examination Workspace (for assessment rules)
- Timetable Workspace (for slots & structure)

7. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Academics
- Academic Setup
- Courses
- Master Courses
- Assessments
- Data Management
- Configuration
- Noticeboard
- What's New

Main Area:

- Table-based management
- Forms for creation/editing
- Calendar views for planning

8. ⚠️ Critical Design Notes

- This workspace must have **strict permissions**
- Changes here affect entire system
- Must support **audit logs**
- Should prevent accidental data corruption

9. 🚀 Why This Workspace is Critical

Without this:

- No class structure
- No subjects
- No exams
- No timetable

👉 It is the **foundation of the entire VID system**

WORKSPACE 2: ADMISSION COUNSELOR WORKSPACE

1. 🧭 Overview

The **Admission Counselor Workspace** is the **entry point of the student lifecycle** in the VID system. It is responsible for handling **all pre-admission activities**, starting from the first enquiry to converting a potential lead into a formal application.

This workspace acts as a **CRM (Customer Relationship Management) layer for the institution**, focused on:

- Lead capture
- Communication tracking
- Conversion management

It ensures that **no enquiry is lost**, and every potential student is properly tracked and nurtured.

2. 👥 Primary Users

- Admission Counselor
- Front Desk Staff
- Marketing / Outreach Team

These users are responsible for **interacting with parents/students** and managing the admission pipeline.

3. 🧩 Complete Module Breakdown

3.1 Enquiry / Lead Management

This is the **first touchpoint module**.

📌 **What it handles:**

- Walk-in enquiries
- Phone enquiries
- Website form submissions
- Campaign-generated leads

⚙️ **Features:**

a) Lead Capture

- Manual entry (front desk input)
- Online form integration
- Bulk import of leads

b) Lead Details Stored

- Student Name
- Parent Name
- Contact Details (Phone, Email)
- Interested Class (e.g., Grade 5)
- Source (Walk-in, Website, Campaign)

- Location

c) Lead Status Tracking

- New
- Contacted
- Follow-up Pending
- Interested
- Not Interested
- Converted

d) Lead Assignment

- Assign leads to specific counselors
- Track ownership

3.2 Communication & Follow-Up Management

This module ensures **continuous engagement with leads**.

⚙️ **Features:**

a) Follow-Up Scheduling

- Set next follow-up date
- Automated reminders for counselors

b) Interaction Logging

- Call logs
- Meeting notes
- Email interactions

c) Communication Channels

- Phone (manual log or integrated)
- Email notifications
- SMS/WhatsApp (optional integration)

d) Status Updates

- Update lead interest level
- Record objections or requirements

3.3 Application Initiation

Once a lead shows interest, the system transitions to **application stage**.

⚙️ **Features:**

a) Convert Lead → Application

- One-click conversion
- Pre-fill data from enquiry

b) Application Form Handling

- Student details
- Parent details
- Previous school details
- Required documents

c) Document Collection

- Upload documents (ID proof, marksheets)
- Track missing documents

3.4 Application Tracking Dashboard

This module provides a **pipeline view of all applications**.

📊 **Views:**

- Pending Applications
- Under Review
- Documents Pending
- Completed Applications

⚙️ **Features:**

- Filter by class, status, date
- Search by student/parent name
- Track conversion rates

3.5 Configuration Module

Used to define **admission rules and structures**.

⚙️ **Features:**

a) Admission Cycle Setup

- Academic year-based admissions
- Application open/close dates

b) Class-wise Availability

- Number of seats per class
- Section capacity visibility

c) Form Customization

- Add/remove fields in application form
- Mandatory vs optional fields

3.6 "What's New" Module

Standard system-wide module.

Purpose:

- Notify counselors about:
  - New features
  - Admission updates

- Process changes

4. 🔄 End-to-End Workflow

1. Capture Enquiry (Walk-in / Online)
2. Assign to Counselor
3. Perform Follow-ups
4. Update Lead Status
5. Convert Interested Lead → Application
6. Collect Documents
7. Forward to Admission Officer

5. 🧠 Data Ownership

This workspace owns:

- Lead / Enquiry Data
- Communication Logs
- Initial Application Data
- Conversion Records

After conversion:
👉 Data is passed to **Admission Officer Workspace**

6. 🔗 Dependencies

Receives Data From:

- Website forms
- Marketing campaigns

Sends Data To:

- Admission Officer Workspace

7. 📊 Key Metrics (Very Important)

This workspace should track:

- Total enquiries
- Conversion rate (Lead → Application)
- Follow-up efficiency
- Counselor performance
- Source-wise performance (which channel brings more students)

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Enquiries / Leads
- Follow-Ups
- Applications
- Configuration
- Reports
- What's New

Main UI:

Dashboard View:

- Total leads
- Today's follow-ups
- Conversion stats

Lead Table:

- Name | Class | Status | Assigned Counselor | Next Follow-up

Pipeline View (Optional):

- Kanban-style (New → Contacted → Converted)

9. ⚠️ Critical Design Notes

- Must support **real-time updates** (leads come anytime)
- Follow-up reminders are **crucial**
- Data duplication must be avoided
- Should support **bulk operations**

10. 🚀 Why This Workspace is Critical

Without this:

- No structured admission pipeline
- Loss of potential students
- No tracking of enquiries

👉 This workspace directly impacts **institution revenue and growth**

WORKSPACE 3: ADMISSION OFFICER WORKSPACE

1. 🧭 Overview

The **Admission Officer Workspace** is responsible for **finalizing the admission process** after a lead has been converted into an application.

While the **Admission Counselor Workspace focuses on lead generation and conversion**, this workspace handles:

- Verification
- Evaluation
- Approval / Rejection
- Final onboarding into the system

It acts as the **decision-making layer of admissions**, ensuring that only eligible students are admitted and properly integrated into the academic system.

2. 👥 Primary Users

- Admission Officer
- Admin Staff
- Principal (for final approval in some cases)

These users have **authority to approve or reject applications** and assign students to the institution.

3. 🧩 Complete Module Breakdown

3.1 Prospects Management

This module contains all **converted leads (applications received)**.

📌 **What it represents:**

- Students who have shown interest and submitted applications
- Awaiting verification or decision

⚙️ **Features:**

a) Prospect Profile

- Student details
- Parent details
- Contact information
- Applied class

b) Status Tracking

- Application Received
- Under Review
- Documents Pending
- Shortlisted
- Rejected
- Approved

c) Filtering & Segmentation

- By class
- By status
- By application date

3.2 Application Verification

This is a **critical validation layer**.

⚙️ **Features:**

a) Document Verification

- Upload and review documents:
  - Birth certificate
  - Previous school records
  - ID proof
- Mark documents as:
  - Verified
  - Rejected
  - Pending

b) Data Validation

- Check for missing or incorrect data
- Ensure mandatory fields are filled

c) Duplicate Check

- Prevent duplicate student entries
- Cross-check with existing records

3.3 Admission Evaluation

This module allows **assessment of applicants**.

⚙️ **Features:**

a) Eligibility Criteria

- Age criteria
- Previous academic performance
- Entrance test (if applicable)

b) Interview Management (Optional)

- Schedule interviews
- Record feedback
- Assign evaluators

c) Scoring System

- Assign scores based on:
  - Academics
  - Interview

    - Other criteria

3.4 Admission Decision Engine

This module handles **final decision-making**.

⚙️ **Features:**

a) Approve / Reject / Waitlist

- Decision options:
    - Approved
    - Rejected
    - Waitlisted

b) Remarks & Justification

- Add comments for decisions
- Maintain audit trail

c) Seat Allocation

- Check available seats in class/section
- Assign section based on capacity

3.5 Student Onboarding (Critical Module)

Once approved, the applicant becomes a **system student**.

⚙️ **Features:**

a) Auto Student Creation

- Convert applicant → student record
- Generate unique Student ID

b) Class & Section Assignment

- Assign to appropriate class and section
- Link to academic structure

c) Initial Setup

- Create login credentials
- Link parent accounts (if enabled)

d) Data Transfer

- Move all application data to Student Workspace

3.6 Admission Reports & Analytics

Provides insights into admission performance.

📊 **Metrics:**

- Total applications
- Approval rate

- Rejection rate
- Class-wise admissions
- Seat occupancy

3.7 Configuration Module

Controls admission-related settings.

⚙️ **Features:**

a) Admission Rules

- Eligibility criteria
- Document requirements

b) Seat Capacity Configuration

- Define max students per class/section

c) Workflow Configuration

- Define approval hierarchy (optional multi-level approval)

3.8 "What's New" Module

Standard update system.

Purpose:

- Notify about:
  - Admission policy updates
  - System feature updates

4. 🔄 End-to-End Workflow

1. Receive Application from Counselor Workspace
2. Verify Documents
3. Validate Data
4. Evaluate Candidate
5. Approve / Reject / Waitlist
6. Assign Class & Section
7. Convert to Student Record
8. Transfer to Student Workspace

5. 🧠 Data Ownership

This workspace owns:

- Application Records
- Verification Status
- Admission Decisions
- Onboarding Data

After onboarding:
👉 Data is transferred to **Student Workspace**

6. 🔗 Dependencies

Receives Data From:

- Admission Counselor Workspace

Sends Data To:

- Student Workspace
- Academic Workspace (for class mapping)

7. 📊 Key Metrics

- Application-to-admission conversion rate
- Approval vs rejection ratio
- Seat utilization per class
- Processing time per application

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Prospects
- Applications
- Verification
- Decisions
- Reports
- Configuration
- What's New

Main UI:

Application Table:

- Name | Class | Status | Documents | Decision

Detail View:

- Full student profile
- Documents section
- Evaluation panel
- Decision buttons

9. ⚠️ Critical Design Notes

- Must ensure **data accuracy before onboarding**
- Should support **audit logs for decisions**
- Seat allocation must be **real-time synced**
- Avoid duplicate student creation

10. 🚀 Why This Workspace is Critical

Without this:

- No structured admission approval
- Risk of invalid or duplicate student records
- No proper onboarding

👉 This workspace ensures **data integrity and controlled admissions**

WORKSPACE 4: ACADEMIC COORDINATOR

1. 🧭 Overview

The **Academic Coordinator (Operational) Workspace** is the **execution layer of academics**.

While the **Academic Admin Workspace defines the structure**, this workspace is responsible for **running and managing day-to-day academic operations** based on that structure.

It acts as a **bridge between planning and execution**, ensuring that:

- Students are correctly enrolled in courses
- Academic progress is tracked
- Faculty activities are aligned
- Academic processes run smoothly

This workspace is **highly dynamic** and used daily.

2. 👥 Primary Users

- Academic Coordinator
- Vice Principal
- Academic Office Staff

These users ensure **academic processes are executed correctly across all classes**.

3. 🧩 Complete Module Breakdown

3.1 Digital Noticeboard

This module is used for **academic-level communication**.

⚙️ Features:

- Publish academic notices (schedule updates, exam alerts)
- Target:
  - Specific classes
  - Sections
  - Entire school
- Attach documents (PDFs, instructions)
- Track read status

3.2 Course Registration

This module handles **student-to-course mapping**.

⚙️ Features:

a) Student Enrollment

- Assign students to courses/subjects
- Bulk enrollment support

b) Course Allocation

- Map courses to classes/sections
- Handle optional/elective subjects (if enabled)

c) Validation

- Ensure no duplicate enrollments
- Check eligibility constraints

3.3 Student Promotion System

Handles **year-end academic transitions**.

⚙️ **Features:**

a) Promotion Logic

- Promote students to next class
- Apply rules (pass/fail criteria)

b) Bulk Promotion

- Promote entire class/section at once

c) Retention Handling

- Handle failed students
- Allow manual overrides

d) Academic Year Transition

- Move students to next academic batch

3.4 All Students Management

Central operational view of all students.

⚙️ **Features:**

- View student list (class-wise, section-wise)
- Filter/search students
- Update student academic details
- Track enrollment status

3.5 Question Bank Management

This module supports **academic content creation**.

⚙️ **Features:**

a) Question Creation

- Add questions by subject
- Define difficulty level
- Tag by topic

b) Categorization

- Subject-wise
- Chapter-wise
- Assessment-wise

c) Reusability

- Use questions in exams and tests
- Create question sets

3.6 Course & Master Course Access

This module consumes data from Academic Admin.

⚙️ **Features:**

- View available courses
- Map courses to students/classes
- Ensure alignment with curriculum

3.7 Attendance Monitoring (Operational Level)

This is a **supervisory layer over attendance**.

⚙️ **Features:**

- View attendance across classes
- Identify irregular students
- Generate attendance reports
- Monitor teacher compliance

3.8 Classroom Management

Handles **physical and logical classroom mapping**.

⚙️ **Features:**

- Assign classrooms to sections
- Manage room capacity
- Track classroom usage

3.9 Reports & Analytics

Provides academic insights.

📊 **Reports Include:**

- Student performance trends
- Attendance summaries
- Class-wise reports
- Subject-wise analysis

⚙️ **Features:**

- Export reports (PDF/Excel)
- Drill-down into student-level data

3.10 Faculty Timetable Oversight

This module ensures **faculty schedules are aligned**.

⚙️ **Features:**

- View faculty timetable
- Identify scheduling conflicts
- Monitor workload distribution

3.11 Academic Activities Tracking

Tracks non-core academic activities.

⚙️ **Features:**

- Record competitions
- Track participation
- Maintain activity history

3.12 "What's New" Module

Standard system update module.

4. 🔄 End-to-End Workflow

1. Receive academic structure from Admin Workspace
2. Enroll students into courses
3. Monitor attendance and activities
4. Manage question banks and assessments
5. Track performance and generate reports
6. Promote students at year-end

5. 🧠 Data Ownership

This workspace manages:

- Student-course mappings
- Promotion records
- Question banks
- Operational academic data

It **does not create core structure**, but operates on it.

6. 🔗 Dependencies

Receives Data From:

- Academic Admin Workspace
- Student Workspace
- Faculty Workspace

Feeds Data To:

- Examination Workspace
- Reports / Dashboard

7. 📊 Key Metrics

- Student promotion rate
- Attendance trends
- Academic performance distribution
- Course enrollment statistics

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Students
- Course Registration
- Promotions
- Question Bank
- Attendance
- Classrooms
- Reports
- Noticeboard
- What's New

Main UI:

Student Table:

- Name | Class | Section | Status

Course Mapping UI:

- Drag-and-drop or selection-based

Reports Dashboard:

- Charts + tables

9. ⚠️ Critical Design Notes

- Must support **bulk operations** (very important)
- Should be **high-performance** (large student data)
- Requires **data consistency with admin workspace**
- Needs **real-time sync with attendance and exams**

10. 🚀 Why This Workspace is Critical

Without this:

- No operational control of academics
- No student progression tracking
- No real academic execution

👉 This workspace ensures that **academic plans actually run in real life**

WORKSPACE 5: STUDENT WORKSPACE

1. 🧭 Overview

The **Student Workspace** is the **primary end-user interface for students**. It provides a **personalized dashboard** where each student can access all academic, administrative, and personal information in one place.

This workspace acts as:

- A **self-service portal**
- A **real-time academic tracker**
- A **communication interface between student and institution**

Unlike admin workspaces, this is **read-heavy with limited write actions**, ensuring students can **view, interact, and respond**, but not alter core data.

2. 👥 Primary Users

- Students
- (Indirectly) Parents via linked accounts

Each student logs in and sees **only their own data**, fully isolated.

3. 🧩 Complete Module Breakdown

3.1 Student Dashboard (Home)

This is the **landing page after login**.

📊 **Displays:**

- Attendance summary (today / monthly %)
- Upcoming classes
- Recent announcements
- Pending tasks (assignments, fees)
- Exam alerts

⚙️ **Features:**

- Quick navigation cards
- Notifications panel
- Personalized greetings

3.2 Profile Management

This module contains **student identity and personal data**.

📌 **Data Includes:**

- Basic details (Name, DOB, Gender)
- Contact information
- Parent/guardian details
- Address
- Documents (ID proof, certificates)

⚙️ **Features:**

- View profile
- Limited edit (based on permissions)
- Upload/update documents (if allowed)

3.3 Attendance Module

Provides **complete visibility of attendance**.

📊 **Views:**

- Daily attendance status
- Monthly summary
- Subject-wise attendance (if enabled)

⚙️ **Features:**

- Visual charts (percentage tracking)
- Absence history
- Alerts for low attendance

3.4 Timetable Module

Displays **student-specific schedule**.

📌 **Includes:**

- Daily/weekly timetable
- Subject, teacher, room details
- Substitution updates

⚙️ **Features:**

- Calendar view
- Real-time updates
- Highlight current/next class

3.5 Examination & Results Module

Handles **academic performance visibility**.

⚙️ **Features:**

a) Exam Schedule

- Upcoming exams
- Subject-wise timetable

b) Results & Report Cards

- Marks per subject
- Grades
- Teacher remarks

c) Performance Insights

- Comparison with class average
- Progress tracking

3.6 Assignments & LMS (Optional but Recommended)

Supports **learning activities**.

⚙️ **Features:**

- View assignments
- Submit homework (file upload)
- Access study materials
- Track submission status

3.7 Fee & Payment Module

Provides **financial transparency**.

📊 **Displays:**

- Fee structure
- Paid amount
- Pending dues

⚙️ **Features:**

- Download receipts
- View transaction history
- Online payment integration (if enabled)

3.8 Co-Curricular & Activity Tracking

Tracks **non-academic participation**.

⚙️ **Features:**

- View events participated
- Achievements & awards
- Club memberships

3.9 Communication Module

Enables **interaction with institution**.

⚙️ **Features:**

- Receive notices & announcements
- Message teachers/admin (if enabled)
- Respond to notifications

3.10 Requests & Services

Allows students to **raise requests**.

Examples:

- Leave request

- Bonafide certificate
- ID card request

⚙️ **Features:**

- Submit request
- Track approval status
- View history

3.11 Documents & Downloads

Central place for **important files**.

⚙️ **Features:**

- Download report cards
- Access certificates
- Store uploaded documents

3.12 Notifications System

Keeps students updated in real-time.

⚙️ **Features:**

- Push notifications
- Alerts for:
  - Exams
  - Fees
  - Attendance
  - Announcements

4. 🔄 End-to-End User Flow

1. Login
2. View Dashboard
3. Check timetable & attendance
4. View assignments/exams
5. Track performance
6. Pay fees or raise requests

5. 🧠 Data Ownership

This workspace **does not create core data**, but consumes from:

- Academic Workspace (classes, timetable)
- Faculty Workspace (attendance, marks)
- Payment Workspace (fees)
- Examination Workspace (results)

It only manages:

- Student interactions
- Requests
- Personal view layer

6. 🔗 Dependencies

Receives Data From:

- Academic Admin
- Faculty
- Examination
- Payment
- Common Workspace

7. 📊 Key Metrics (Student-Level)

- Attendance percentage
- Academic performance
- Assignment completion rate
- Fee payment status

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Profile
- Attendance
- Timetable
- Exams & Results
- Assignments
- Fees
- Activities
- Requests
- Documents

Main UI:

Dashboard Cards:

- Attendance %
- Upcoming Exams
- Pending Fees

Charts:

- Attendance graph
- Performance trend

9. ⚠️ Critical Design Notes

- Must be **mobile-first (students mostly use phones)**
- UI should be **simple and intuitive**
- Data must be **strictly isolated per student**
- Should support **real-time updates**

10. 🚀 Why This Workspace is Critical

Without this:

- No student visibility
- No engagement
- No self-service

👉 This workspace is the **face of the entire system for students**

WORKSPACE 6: FACULTY WORKSPACE

1. 🧭 Overview

The **Faculty Workspace** is the **primary operational interface for teachers**. It enables faculty members to **execute all teaching-related responsibilities**, including:

- Conducting classes
- Marking attendance
- Managing assignments
- Evaluating students
- Communicating with students

This workspace is **action-heavy** (unlike student workspace which is view-heavy) and plays a **critical role in generating academic data** used across the system.

It acts as:

- The **execution layer of teaching**
- The **data input source** for attendance and performance
- The **interaction layer between teacher and students**

2. 👥 Primary Users

- Teachers / Faculty
- Class Teachers
- Subject Teachers

Each faculty member sees:

- Only their assigned classes
- Their subjects
- Their timetable

3. 🧩 Complete Module Breakdown

3.1 Faculty Dashboard (Home)

The central control panel for teachers.

📊 **Displays:**

- Today's timetable
- Pending tasks (attendance, grading)
- Recent announcements
- Upcoming classes

⚙️ **Features:**

- Quick access to classes
- Notifications panel
- Task reminders

3.2 My Classes / Courses

This module shows **all assigned teaching responsibilities**.

📌 **Includes:**

- Class name (e.g., Grade 5 – Section A)
- Subject assigned
- Student list

⚙️ **Features:**

- View class roster
- Access student profiles
- Navigate to attendance / assignments

3.3 Attendance Management

One of the **most critical modules**.

⚙️ **Features:**

a) Daily Attendance

- Mark present/absent
- Period-wise or day-wise attendance

b) Bulk Operations

- Mark attendance for entire class quickly

c) Edit & Correction

- Update attendance (with permissions)

d) History Tracking

- View past attendance records

3.4 Assignment & Homework Management

Handles **academic task distribution**.

⚙️ **Features:**

a) Assignment Creation

- Create homework/classwork
- Add instructions
- Attach files

b) Submission Tracking

- View student submissions
- Track pending work

c) Evaluation

- Grade assignments
- Provide feedback

3.5 Assessment & Marks Entry

Feeds data into the **Examination system**.

⚙️ **Features:**

a) Marks Entry

- Enter marks for exams/tests
- Subject-wise entry

b) Internal Assessments

- Continuous evaluation (projects, activities)

c) Validation

- Ensure marks within limits
- Prevent errors

3.6 Question Bank Access

Used for **content creation and reuse**.

⚙️ **Features:**

- Create new questions
- Categorize by topic/difficulty
- Reuse for tests and exams

3.7 Lesson Planning (Recommended)

Helps structure teaching.

⚙️ **Features:**

- Plan lessons by topic
- Track syllabus completion
- Align with academic calendar

3.8 Timetable View

Displays **teacher-specific schedule**.

📌 **Includes:**

- Class timings
- Subject details
- Room info

⚙️ **Features:**

- Daily/weekly view
- Substitution updates

3.9 Student Performance Tracking

Provides insights into **individual student progress**.

⚙️ **Features:**

- View student marks
- Add remarks
- Identify weak students
- Track improvement

3.10 Communication Module

Enables **interaction with students and parents**.

⚙️ **Features:**

- Send announcements to class
- Share updates
- Respond to student queries

3.11 Reports & Analytics

Provides **teaching-related insights**.

📊 **Reports:**

- Attendance reports
- Marks distribution
- Class performance

⚙️ **Features:**

- Export reports
- Analyze trends

3.12 Classroom / Session Management (Optional Advanced)

For advanced setups.

⚙️ **Features:**

- Track live sessions
- Record class notes
- Manage virtual classes (if integrated)

3.13 Notifications & "What's New"

Keeps faculty updated.

4. 🔄 End-to-End Workflow

1. Login
2. View timetable
3. Conduct class
4. Mark attendance
5. Assign homework
6. Evaluate submissions
7. Enter marks
8. Track student performance

5. 🧠 Data Ownership

This workspace **generates critical academic data**:

- Attendance records
- Assignment data
- Marks and evaluations
- Teacher remarks

This data feeds into:
👉 Student Workspace
👉 Examination Workspace
👉 Reports

6. 🔗 Dependencies

Receives Data From:

- Academic Workspace (classes, subjects)
- Timetable Workspace

Sends Data To:

- Student Workspace
- Examination Workspace
- Reports/Dashboard

7. 📊 Key Metrics

- Attendance completion rate
- Assignment completion rate
- Student performance trends
- Teacher workload

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- My Classes
- Attendance
- Assignments
- Assessments
- Question Bank
- Timetable
- Reports
- Communication

Main UI:

Class View:

- Student list with actions

Attendance Screen:

- Checkbox/grid-based marking

Assignment Panel:

- Create + track submissions

9. ⚠️ Critical Design Notes

- Must be **fast and simple (used daily)**
- Attendance marking should be **1-click efficient**
- Avoid complex UI — teachers need speed
- Should support **mobile/tablet usage**

10. 🚀 Why This Workspace is Critical

Without this:

- No attendance data
- No academic performance data
- No teaching execution

👉 This workspace is the **engine that generates academic data for the entire system**

WORKSPACE 7: EXAMINATION WORKSPACE

1. 🧭 Overview

The **Examination Workspace** is the **central evaluation engine** of the VID system.

It manages the **complete lifecycle of assessments**, including:

- Exam planning
- Scheduling
- Marks collection
- Result processing
- Report card generation
- Performance analytics

This workspace transforms **raw academic inputs (marks)** into **meaningful outputs (results, grades, insights)**.

It acts as:

- The **academic validation layer**
- The **result publishing system**
- The **performance analytics engine**

2. 👥 Primary Users

- Examination Controller
- Academic Coordinator
- Faculty (Marks Entry)
- Principal (Monitoring & Insights)

3. 🧩 Complete Module Breakdown

3.1 Examination Dashboard

Central control panel for exams.

📊 **Displays:**

- Upcoming exams
- Marks entry status
- Result publishing progress
- Alerts (pending marks, errors)

⚙️ **Features:**

- Quick navigation to exams
- Status indicators (draft / active / completed)

3.2 Exam Configuration (Exam Setup)

Defines the **structure of exams**.

⚙️ **Features:**

a) Exam Creation

- Define exam name (Midterm, Final, Unit Test)
- Select academic year

b) Exam Type

- Theory / Practical / Internal

c) Subject Mapping

- Assign subjects to exam

d) Weightage Setup

- Marks distribution
- Internal vs external weightage

3.3 Exam Scheduling

Handles **when exams occur**.

⚙️ **Features:**

- Create subject-wise exam timetable
- Assign:
  - Date
  - Time
  - Duration
- Avoid conflicts

📌 **Output:**

- Student exam timetable
- Faculty invigilation schedule (optional)

3.4 Marks Entry Management

This is the **core data collection module**.

⚙️ **Features:**

a) Faculty Input

- Teachers enter marks per subject

b) Bulk Upload

- Upload marks via Excel

c) Validation

- Check max marks
- Prevent invalid entries

d) Draft & Submit Flow

- Save draft
- Final submission lock

3.5 Marks Verification & Moderation

Ensures **accuracy and fairness**.

⚙️ **Features:**

- Review submitted marks
- Approve/reject entries
- Apply moderation rules
- Adjust marks if needed

3.6 Result Processing Engine

Converts marks into **final results**.

⚙️ **Features:**

a) Grade Calculation

- Apply grading rules
- Convert marks to grades

b) Aggregation

- Combine multiple assessments

c) Pass/Fail Logic

- Based on rules defined in Academic Admin

3.7 Report Card Generation

Creates **student report cards**.

⚙️ **Features:**

- Generate report cards automatically
- Include:
  - Subject-wise marks
  - Grades
  - Remarks
- Support templates (customizable design)

3.8 Result Publishing

Controls **when results are visible**.

⚙️ **Features:**

- Publish results to students/parents
- Role-based visibility
- Schedule release

3.9 Performance Analytics

Provides deep academic insights.

📊 **Analytics Include:**

- Class average
- Subject-wise performance
- Top performers
- Weak students

⚙️ **Features:**

- Charts and graphs
- Trend analysis
- Comparative reports

3.10 Previous Papers & Records

Stores historical exam data.

⚙️ **Features:**

- Access past papers
- View previous results
- Archive records

3.11 Exam Reports

Provides administrative reporting.

📊 **Reports:**

- Marks reports
- Result summaries
- Performance analysis

3.12 Notifications & Alerts

Keeps stakeholders informed.

⚙️ **Alerts:**

- Marks pending
- Results published
- Exam schedule updates

4. 🔄 End-to-End Workflow

1. Create exam structure
2. Schedule exams
3. Faculty enters marks
4. Verify & validate marks
5. Process results
6. Generate report cards
7. Publish results
8. Analyze performance

5. 🧠 Data Ownership

This workspace manages:

- Exam structures
- Marks data
- Grades and results
- Report cards

It transforms:
👉 Faculty input → Final student results

6. 🔗 Dependencies

Receives Data From:

- Academic Workspace (subjects, structure)
- Faculty Workspace (marks)

Sends Data To:

- Student Workspace (results)
- Parent Portal
- Principal Dashboard

7. 📊 Key Metrics

- Pass percentage
- Class average
- Subject performance
- Topper analysis
- Failure trends

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Exams
- Schedule
- Marks Entry
- Verification
- Results
- Report Cards
- Analytics
- Reports

Main UI:

Marks Entry Screen:

- Table format (student vs marks)

Analytics Dashboard:

- Graphs (bar, line charts)

9. ⚠️ Critical Design Notes

- Must ensure **data accuracy (zero tolerance for errors)**

- Strong **validation and locking mechanisms**
- Should support **bulk operations**
- Must handle **large datasets efficiently**

10. 🚀 Why This Workspace is Critical

Without this:

- No exams
- No results
- No academic validation

👉 This workspace is the **final authority on student performance**

WORKSPACE 8: ADMISSION OFFICER WORKSPACE

1. 🧭 Overview

The **Admission Officer Workspace** is the **complete student onboarding pipeline system**.

It manages the journey from:
👉 **Enquiry → Application → Verification → Admission → Student Creation**

This workspace is critical because it is the **entry point of all student data** into the VID system.

It acts as:

- A **pipeline management system (like CRM for admissions)**
- A **decision-making tool for admissions**
- A **data creation layer for student records**

2. 👥 Primary Users

- Admission Officer
- Front Desk / Enquiry Team
- Admission Manager
- Admin (for approvals)

3. 🧩 Complete Module Breakdown

3.1 Admission Dashboard

Central view of the **admission pipeline**.

📊 **Displays:**

- Total enquiries
- Applications in progress
- Approved admissions
- Rejected applications
- Conversion rate (enquiry → admission)

⚙️ **Features:**

- Funnel visualization (pipeline stages)
- Quick filters (class-wise, date-wise)

3.2 Prospect / Enquiry Management

This is the **first stage of admissions**.

⚙️ **Features:**

a) Enquiry Capture

- Walk-in enquiries
- Phone enquiries
- Website leads

b) Prospect Profile

- Student basic details
- Parent contact info
- Interested class/grade

c) Follow-Up Tracking

- Call logs
- Visit history
- Follow-up reminders

3.3 Application Management

Handles **formal admission applications**.

⚙️ **Features:**

a) Application Form

- Online/offline form submission
- Custom fields (based on institution)

b) Document Upload

- Birth certificate
- Previous school records
- ID proof

c) Status Tracking

- Draft
- Submitted
- Under Review

3.4 Admission Workflow Engine

Controls **decision-making flow**.

⚙️ **Features:**

a) Multi-Step Approval

- Screening
- Interview (optional)
- Final approval

b) Eligibility Checks

- Age criteria
- Academic performance

c) Status Transitions

- Applied → Reviewed → Approved/Rejected

3.5 Admission Decision & Approval

Final stage before onboarding.

⚙️ **Features:**

- Approve/reject applications
- Add remarks
- Notify parents/students

3.6 Student Onboarding (Critical Module)

Converts approved applicants into **system students**.

⚙️ **Features:**

a) Student Record Creation

- Generate student ID (VID)
- Create profile in Student Workspace

b) Class Allocation

- Assign class & section

c) Fee Setup

- Link fee structure

3.7 Document Verification

Ensures **authenticity of student data**.

⚙️ **Features:**

- Verify uploaded documents
- Mark verified/unverified
- Flag discrepancies

3.8 Admission Reports

Provides insights into admissions.

📊 **Reports:**

- Enquiry vs admission conversion
- Class-wise admissions
- Source-wise leads (online, walk-in)

3.9 Communication Module

Handles communication with prospects/parents.

⚙️ **Features:**

- Send SMS/email updates
- Notify status changes
- Share admission instructions

3.10 Configuration Module

Controls admission settings.

⚙️ **Features:**

- Define admission cycles
- Configure application forms
- Set eligibility rules

3.11 "What's New" Module

System updates and feature announcements.

4. 🔄 End-to-End Workflow

1. Capture enquiry
2. Create prospect profile
3. Submit application
4. Upload documents
5. Verify & review
6. Approve/reject application
7. Convert to student
8. Assign class & fees

5. 🧠 Data Ownership

This workspace creates:

- Prospect data
- Application records
- Admission decisions
- Initial student profiles

👉 It is the **source of truth for student onboarding**

6. 🔗 Dependencies

Sends Data To:

- Student Workspace (new student creation)
- Payment Workspace (fee setup)
- Academic Workspace (class allocation)

Receives From:

- Website/lead sources
- Admin configurations

7. 📊 Key Metrics

- Conversion rate (enquiry → admission)
- Application processing time
- Admission trends
- Source performance

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Prospects
- Applications
- Admissions
- Onboarding
- Documents
- Reports
- Configuration

Main UI:

Pipeline View:

- Enquiry → Application → Approval

Application Table:

- Name | Class | Status | Action

9. ⚠️ Critical Design Notes

- Must support **high volume during admission season**
- Requires **pipeline tracking (CRM-like UX)**
- Must ensure **data accuracy (student creation is critical)**
- Needs **automation (notifications, status updates)**

10. 🚀 Why This Workspace is Critical

Without this:

- No student onboarding
- No structured admission process
- No data entry into system

👉 This workspace is the **gateway into the entire VID system**

WORKSPACE 9: PAYMENT ADMINISTRATION WORKSPACE

1. 🧭 Overview

The **Payment Administration Workspace** is the **financial backbone of the VID system**.

It manages all **student-related financial operations**, including:

- Fee structure setup
- Fee collection (online/offline)
- Concessions & scholarships
- Financial tracking & reporting
- Payment reconciliation

This workspace is **highly sensitive and critical**, as it deals with **money, compliance, and accounting accuracy**.

It acts as:

- A **fee management engine**
- A **transaction processing system**
- A **financial reporting module**

2. 👥 Primary Users

- Accounts Team
- Cashier
- Finance Admin
- Institution Admin (Oversight)

3. 🧩 Complete Module Breakdown

3.1 Financial Dashboard

Central overview of financial health.

📊 **Displays:**

- Total fees collected
- Pending dues
- Daily transactions
- Revenue trends

⚙️ **Features:**

- Filters (class-wise, date-wise)
- Alerts for defaulters

3.2 Fee Structure Management

Defines **how fees are organized**.

⚙️ **Features:**

a) Fee Heads

- Tuition fee
- Transport fee
- Exam fee
- Miscellaneous charges

b) Installments

- Term-wise or monthly fee setup

c) Class-wise Mapping

- Different fees for different classes

3.3 Student Fee Mapping

Links **fee structures to students**.

⚙️ **Features:**

- Assign fee plans to students
- Bulk assignment
- Modify fee structure per student

3.4 Academic Fees Module

Handles **regular school fees**.

⚙️ **Features:**

- Generate fee demands
- Track payments
- Manage dues

3.5 Examination Fees Module

Handles **exam-related charges**.

⚙️ **Features:**

- Define exam fee structure
- Collect exam payments
- Restrict hall ticket if unpaid (optional)

3.6 Online Transactions

Handles **digital payments**.

⚙️ **Features:**

- Payment gateway integration
- Real-time payment status
- Transaction tracking

3.7 Offline Payments (Counter Payments)

Handles **manual payments**.

⚙️ **Features:**

- Cash/cheque entry
- Generate receipts
- Update ledger

3.8 Receipt & Invoice Management

Generates **financial documents**.

⚙️ **Features:**

- Auto-generate receipts
- Download/print invoices
- Reprint receipts

3.9 Concessions & Discounts

Manages **fee reductions**.

⚙️ **Features:**

- Apply concessions
- Define discount rules
- Approve/reject concessions

3.10 Scholarships Management

Handles **financial aid programs**.

⚙️ **Features:**

- Assign scholarships
- Track eligibility
- Apply reductions automatically

3.11 Challan Management

Used for **bank-based payments**.

⚙️ **Features:**

- Generate challans
- Track bank payments
- Reconcile with system

3.12 Settlements & Reconciliation

Ensures **financial accuracy**.

⚙️ **Features:**

- Match transactions with bank data
- Identify mismatches
- Resolve discrepancies

3.13 Credit Memo & Adjustments

Handles **refunds and corrections**.

⚙️ **Features:**

- Issue refunds
- Adjust excess payments
- Maintain audit trail

3.14 Student Fee Card

Provides **complete fee history per student**.

📊 **Includes:**

- Total fee
- Paid amount
- Pending dues
- Transaction history

3.15 Payment Permissions & Restrictions

Controls **who can pay what**.

⚙️ **Features:**

- Restrict exam access if fees unpaid
- Enable/disable payment modules
- Control fee visibility

3.16 Reports & Financial Analytics

Provides **financial insights**.

📊 **Reports:**

- Collection reports
- Defaulter lists
- Transaction reports
- Class-wise revenue

3.17 Open Payments Module

Handles **miscellaneous payments**.

⚙️ **Features:**

- Accept ad-hoc payments
- Link to student or general account

3.18 Configuration Module

Controls system-level financial settings.

⚙️ **Features:**

- Payment gateway setup
- Currency settings
- Tax configuration (if applicable)

4. 🔄 End-to-End Workflow

1. Define fee structure
2. Assign fees to students
3. Generate fee demands
4. Collect payments (online/offline)
5. Issue receipts
6. Track dues
7. Apply concessions/scholarships
8. Reconcile transactions
9. Generate reports

5. 🧠 Data Ownership

This workspace manages:

- Fee structures
- Payment transactions
- Receipts and invoices
- Financial adjustments

👉 It is the **single source of truth for financial data**

6. 🔗 Dependencies

Receives Data From:

- Admission Workspace (new students)
- Academic Workspace (class mapping)

Sends Data To:

- Student Workspace (fee visibility)
- Parent Portal (payments)
- Admin Dashboard (financial analytics)

7. 📊 Key Metrics

- Total revenue
- Collection efficiency (%)
- Outstanding dues
- Payment trends
- Scholarship impact

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Fee Structure
- Student Fees
- Transactions
- Receipts
- Concessions
- Scholarships

- Reports
- Configuration

Main UI:

Transaction Table:

- Student | Amount | Mode | Status

Dashboard Charts:

- Revenue trends
- Dues breakdown

9. ⚠️ Critical Design Notes

- Must ensure **100% accuracy (financial data is sensitive)**
- Strong **audit trail required**
- Must support **bulk operations**
- Needs **secure payment integrations**
- Role-based access is **mandatory**

10. 🚀 Why This Workspace is Critical

Without this:

- No fee management
- No revenue tracking
- No financial control

👉 This workspace is the **financial engine of the institution**

WORKSPACE 10: HRMS (HUMAN RESOURCE MANAGEMENT SYSTEM)

1. 🧭 Overview

The **HRMS Workspace** is the **complete employee lifecycle management system** within VID.

It manages everything related to **staff and employees**, including:

- Employee onboarding
- Attendance & leave tracking
- Payroll processing
- Performance management
- Staff records & compliance

This workspace ensures that **all human resources in the institution are managed efficiently and transparently**.

It acts as:

- A **staff database system**
- A **payroll engine**
- A **workforce management tool**

2. 👥 Primary Users

- HR Team
- Admin / Principal
- Accounts Team (Payroll)
- Employees (Self-service access)

3. 🧩 Complete Module Breakdown

3.1 HR Dashboard

Central overview of workforce data.

📊 **Displays:**

- Total employees
- Attendance summary
- Leave statistics
- Payroll status

⚙️ **Features:**

- Alerts (absent staff, pending approvals)
- Workforce insights

3.2 Employee Master (Core Module)

This is the **central employee database**.

📌 **Data Includes:**

- Personal details (Name, DOB, Gender)
- Contact details
- Job role & department
- Joining date
- Documents (ID proof, certificates)

⚙️ **Features:**

- Add/edit employee
- Upload documents
- Assign roles

3.3 Attendance Management

Tracks **employee attendance**.

⚙️ **Features:**

a) Daily Attendance

- Mark attendance
- Biometric/QR integration (optional)

b) Attendance Logs

- Check-in/check-out times

c) Reports

- Monthly attendance summary

3.4 Leave Management

Handles **employee leave requests**.

⚙️ **Features:**

a) Leave Types

- Casual leave
- Sick leave
- Paid leave

b) Leave Requests

- Apply for leave
- Approval workflow

c) Leave Balance

- Track remaining leaves

3.5 Payroll Management (Critical Module)

Handles **salary processing**.

⚙️ **Features:**

a) Salary Structure

- Basic salary
- Allowances
- Deductions

b) Payroll Processing

- Monthly salary calculation
- Auto deductions (leave, tax if applicable)

c) Payslips

- Generate payslips
- Download/print

3.6 Expenses & Reimbursements

Tracks **employee expenses**.

⚙️ **Features:**

- Submit expense claims
- Approve/reject claims
- Reimbursement tracking

3.7 Shift & Work Schedule Management

Manages **employee schedules**.

⚙️ **Features:**

- Define shifts
- Assign shifts
- Handle shift changes

3.8 Compensatory Leave Management

Handles **extra work compensation**.

⚙️ **Features:**

- Apply for compensatory leave
- Track approvals
- Link to overtime

3.9 Performance Management

Tracks **employee performance**.

⚙️ **Features:**

- Performance reviews
- Feedback cycles
- Appraisal tracking

3.10 Employee Self-Service Portal

Allows employees to **manage their own data**.

⚙️ **Features:**

- View profile
- Apply for leave
- Download payslips
- Track attendance

3.11 HR Reports & Analytics

Provides workforce insights.

📊 **Reports:**

- Attendance reports
- Leave reports
- Payroll reports
- Employee performance reports

3.12 Document Management

Handles employee-related documents.

⚙️ **Features:**

- Upload/store documents
- Secure access
- Version tracking

3.13 Notifications & Alerts

Keeps staff informed.

⚙️ **Features:**

- Leave approval alerts
- Payroll notifications
- Attendance alerts

3.14 Configuration Module

Controls HR settings.

⚙️ **Features:**

- Define leave policies
- Configure salary rules
- Set attendance rules

4. 🔄 End-to-End Workflow

1. Add employee
2. Assign role & department
3. Track attendance

4. Manage leaves
5. Process payroll
6. Generate payslips
7. Track performance

5. 🧠 Data Ownership

This workspace manages:

- Employee records
- Attendance logs
- Leave data
- Payroll data

👉 It is the **single source of truth for staff data**

6. 🔗 Dependencies

Receives Data From:

- User Management System (roles)

Sends Data To:

- Payment Workspace (salary processing if integrated)
- Admin Dashboard (workforce analytics)

7. 📊 Key Metrics

- Attendance rate
- Leave utilization
- Payroll expenses
- Staff performance

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Employees
- Attendance
- Leaves
- Payroll
- Expenses
- Reports
- Configuration

Main UI:

Employee Table:

- Name | Role | Department | Status

Payroll Screen:

- Salary breakdown

9. ⚠️ Critical Design Notes

- Must ensure **data privacy (employee data is sensitive)**
- Payroll must be **accurate and auditable**
- Needs **approval workflows**
- Should support **self-service access**

10. 🚀 Why This Workspace is Critical

Without this:

- No staff management
- No payroll system
- No HR tracking

👉 This workspace is the **people management engine of the institution**

WORKSPACE 11: TIMETABLE WORKSPACE

1. 🧭 Overview

The **Timetable Workspace** is the **central scheduling engine** of the VID system.

It is responsible for creating and managing:

- Class schedules
- Faculty schedules
- Room allocations
- Substitutions

This workspace ensures that **academic activities are structured, conflict-free, and optimized**.

It acts as:

- A **time allocation engine**
- A **resource management system**
- A **dependency hub connecting students, faculty, and classrooms**

2. 👥 Primary Users

- Timetable Coordinator
- Academic Coordinator
- Admin / Principal (Oversight)
- Faculty (View access)
- Students (View access)

3. 🧩 Complete Module Breakdown

3.1 Timetable Dashboard

Central control panel for scheduling.

📊 **Displays:**

- Active timetables
- Pending schedules
- Conflicts (if any)
- Substitution alerts

⚙️ **Features:**

- Quick access to classes/faculty schedules
- Conflict notifications

3.2 Academic Time Structure Setup

Defines the **foundation of scheduling**.

⚙️ **Features:**

a) Period Configuration

- Number of periods per day
- Duration of each period

b) Working Days

- Define weekdays
- Holidays integration

c) Breaks

- Lunch breaks
- Short breaks

3.3 Class Timetable Generation

Creates **section-wise schedules**.

⚙️ **Features:**

a) Manual Scheduling

- Assign subject to each period

b) Auto-Generation (Advanced)

- Generate timetable using constraints

c) Constraints Handling

- Subject frequency per week
- Teacher availability
- Room availability

3.4 Faculty Timetable Management

Handles **teacher schedules**.

⚙️ **Features:**

- View teacher timetable
- Ensure no double-booking
- Balance workload

3.5 Subject-Period Allocation

Defines **how often subjects are taught**.

⚙️ **Features:**

- Assign number of periods per subject
- Weekly distribution
- Priority subjects

3.6 Classroom & Resource Allocation

Manages **physical resources**.

⚙️ **Features:**

- Assign classrooms
- Manage lab schedules
- Track room usage

3.7 Substitution Management

Handles **real-time schedule changes**.

⚙️ **Features:**

a) Substitute Teacher Assignment

- Replace absent teacher

b) Temporary Adjustments

- Swap periods

c) Notifications

- Inform students and faculty

3.8 Timetable Publishing

Controls **visibility of schedules**.

⚙️ **Features:**

- Publish timetable
- Version control
- Schedule updates

3.9 Timetable Views

Provides multiple perspectives.

📊 **Views:**

- Class-wise timetable
- Faculty-wise timetable
- Room-wise timetable

3.10 Timetable Reports

Provides scheduling insights.

📊 **Reports:**

- Teacher workload
- Room utilization
- Period distribution

3.11 Notifications & Alerts

Keeps stakeholders informed.

⚙️ **Alerts:**

- Timetable updates
- Substitution notifications
- Conflict alerts

3.12 Configuration Module

Controls scheduling rules.

⚙️ **Features:**

- Define constraints
- Set scheduling priorities
- Configure auto-generation rules

4. 🔄 End-to-End Workflow

1. Define academic time structure
2. Configure subjects & constraints
3. Generate timetable
4. Assign classrooms
5. Publish timetable
6. Handle substitutions
7. Monitor and update

5. 🧠 Data Ownership

This workspace manages:

- Timetable data
- Period allocations
- Scheduling rules
- Substitution logs

6. 🔗 Dependencies

Receives Data From:

- Academic Workspace (classes, subjects)
- Faculty Workspace (teacher availability)

Sends Data To:

- Student Workspace (student timetable)
- Faculty Workspace (teacher timetable)
- Admin Dashboard

7. 📊 Key Metrics

- Teacher workload balance
- Room utilization efficiency
- Schedule conflicts
- Period distribution

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Timetables
- Classes
- Faculty
- Substitutions
- Reports
- Configuration

Main UI:

Grid View (Core UI):

- Rows = Periods
- Columns = Days

Drag & Drop Scheduling:

- Assign subjects easily

9. ⚠️ Critical Design Notes

- Must handle **complex constraints efficiently**
- Needs **conflict detection system**
- Should support **auto + manual scheduling**
- Must allow **quick real-time changes**

10. 🚀 Why This Workspace is Critical

Without this:

- No structured academic schedule
- Conflicts in classes and faculty
- Inefficient resource usage

👉 This workspace is the **time management engine of the institution**

WORKSPACE 12: COMMON WORKSPACE (STUDENT CENTRAL)

1. 🧭 Overview

The **Common Workspace (Student Central)** is the **shared services layer** of the VID system.

It acts as a **central hub** that connects multiple workspaces and provides:

- Cross-functional access
- Unified services
- Shared communication channels
- Request & approval systems

Unlike other workspaces that are **domain-specific (academics, payments, etc.)**, this workspace is **cross-domain**, meaning it integrates and exposes features from multiple modules in a **single unified interface**.

It acts as:

- A **centralized service layer**
- A **user interaction hub**
- A **cross-workspace integration interface**

2. 👥 Primary Users

- Students
- Faculty
- Admin Staff
- Parents (if enabled)

👉 This is one of the **most widely used workspaces across all roles**.

3. 🧩 Complete Module Breakdown

3.1 Home Dashboard

Central landing page for all users.

📊 **Displays:**

- Announcements
- Pending requests
- Notifications
- Quick links to services

⚙️ **Features:**

- Personalized view (based on role)
- Unified access to modules

3.2 Student Central (Core Module)

Acts as a **unified access layer for student-related services**.

⚙️ **Features:**

- View student-related data
- Access academic, payment, and exam info
- Centralized student profile view

3.3 Notice Board (Digital Noticeboard)

One of the **most important shared modules**.

⚙️ **Features:**

- Publish notices
- Target specific groups (class, role)
- Attach documents
- Track read acknowledgments

3.4 Requests & Service Management

Handles **user-generated requests**.

⚙️ **Features:**

a) Request Types

- Leave requests
- Certificate requests
- Permission requests

b) Workflow

- Submit → Review → Approve/Reject

c) Tracking

- Status updates
- Request history

3.5 My Requests

Personal tracking module.

⚙️ **Features:**

- View submitted requests
- Check status
- Re-submit if needed

3.6 Payments (Linked Module)

Provides **quick access to financial data**.

⚙️ **Features:**

- View fee details
- Check dues
- Access payment history

👉 Data comes from Payment Workspace.

3.7 Examinations (Linked Module)

Provides **quick exam-related access**.

⚙️ **Features:**

- View results
- Access exam schedule
- Download report cards

👉 Data comes from Examination Workspace.

3.8 HRMS Access (Linked Module)

Provides **employee-related services**.

⚙️ **Features:**

- Apply for leave
- View attendance
- Access payroll data

👉 Data comes from HRMS Workspace.

3.9 User Management System (UMS Access)

Integrated access to identity management.

⚙️ **Features:**

- View user roles
- Manage permissions (for admins)
- Profile access

3.10 Communication Module

Handles **general communication**.

⚙️ **Features:**

- Messaging
- Broadcast announcements
- Notifications

3.11 Configuration Module

Controls shared settings.

⚙️ **Features:**

- Configure request workflows
- Manage notice settings
- Define access permissions

3.12 Notifications & Alerts

Central notification system.

⚙️ **Features:**

- Real-time alerts
- Multi-module notifications
- Priority alerts

3.13 “What’s New” Module

Platform-wide feature updates.

4. 🔄 End-to-End Workflow

1. User logs in
2. Accesses Home Dashboard
3. Views notices and updates
4. Raises requests
5. Tracks request status
6. Accesses linked modules (fees, exams, HRMS)

5. 🧠 Data Ownership

This workspace **does not own core data**, but manages:

- Requests
- Notices
- Communication logs

It acts as a:
👉 **Data aggregator + service layer**

6. 🔗 Dependencies

Receives Data From:

- Payment Workspace
- Examination Workspace
- HRMS Workspace
- Student Workspace

Acts As Interface For:

- All users across system

7. 📊 Key Metrics

- Number of requests raised
- Request resolution time
- Notice engagement rate
- User activity

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Home

- Student Central
- Notice Board
- Requests
- My Requests
- Payments
- Exams
- HRMS
- Configuration

Main UI:

Dashboard Cards:

- Pending Requests
- Notifications

Request Panel:

- Submit + Track

9. ⚠️ Critical Design Notes

- Must be **highly responsive (used by all users)**
- Needs **clean and simple UI**
- Should support **cross-module navigation**
- Must ensure **role-based visibility**

10. 🚀 Why This Workspace is Critical

Without this:

- No centralized user interaction
- No request system
- No unified experience

👉 This workspace is the **central interaction hub of the entire VID platform**

WORKSPACE 13: PRINCIPAL / ADMIN DASHBOARD WORKSPACE

1. 🧭 Overview

The **Principal / Admin Dashboard Workspace** is the **strategic control center** of the entire VID system.

It provides:

- Institution-wide visibility
- Real-time analytics
- Performance insights
- Decision-making tools

This workspace **does not manage raw data directly**. Instead, it:
👉 Aggregates data from all workspaces
👉 Converts it into meaningful insights

It acts as:

- A **decision intelligence system**
- A **monitoring and control panel**
- A **performance analytics hub**

2. 👥 Primary Users

- Principal
- Institution Admin
- Management Team
- Leadership Stakeholders

3. 🧩 Complete Module Breakdown

3.1 Executive Dashboard (Core Module)

The main interface showing **institution health at a glance**.

📊 **Key Metrics Displayed:**

Academic Metrics

- Student performance trends
- Pass percentage
- Subject-wise performance

Attendance Metrics

- Student attendance rate
- Faculty attendance rate

Financial Metrics

- Fee collection status
- Pending dues

Operational Metrics

- Active classes
- Daily activities

3.2 Academic Performance Analytics

Deep insights into academics.

⚙️ **Features:**

- Class-wise performance
- Subject analysis
- Top performers
- Weak areas identification

📊 **Outputs:**

- Graphs
- Trends
- Comparative analysis

3.3 Attendance Analytics

Tracks attendance patterns.

⚙️ **Features:**

- Daily attendance overview
- Monthly trends
- Absentee alerts

📊 **Insights:**

- Low attendance detection
- Class-wise attendance patterns

3.4 Financial Analytics

Provides financial overview.

⚙️ **Features:**

- Fee collection dashboard
- Pending dues analysis
- Payment trends

📊 **Insights:**

- Revenue trends
- Defaulter identification

3.5 Operational Monitoring

Tracks day-to-day activities.

⚙️ **Features:**

- Pending approvals
- Ongoing processes
- System alerts

3.6 Workforce Insights (HRMS Integration)

Tracks staff performance.

⚙️ **Features:**

- Teacher performance
- Attendance of staff
- Workload analysis

3.7 Alerts & Exception Management

Highlights critical issues.

⚙️ **Alerts:**

- Low attendance
- Fee defaulters
- Academic underperformance
- Pending approvals

3.8 Reports Center

Central repository of reports.

📊 **Reports:**

- Academic reports
- Financial reports
- Attendance reports
- Operational reports

⚙️ **Features:**

- Download/export
- Filter-based reports

3.9 Drill-Down Analysis

Allows deeper investigation.

⚙️ **Features:**

- Click → drill into detailed data
- Navigate to source workspace

Example:
👉 Click “Low Attendance” → opens Student Workspace data

3.10 Multi-Workspace Integration View

Shows combined insights.

⚙️ **Features:**

- Combine data across modules
- Cross-functional analytics

3.11 Notifications & Decision Support

Supports leadership actions.

⚙️ **Features:**

- Real-time alerts
- Suggested actions (AI-ready)

3.12 Custom Dashboard Builder (Advanced)

Allows customization.

⚙️ **Features:**

- Add/remove widgets
- Personalize dashboard

4. 🔄 End-to-End Workflow

1. Admin logs in
2. Views dashboard summary
3. Identifies issues (alerts)
4. Drills down into data
5. Takes action via respective workspace
6. Monitors improvements

5. 🧠 Data Ownership

This workspace:

- **Does not own primary data**

Instead, it:
👉 Aggregates data from all workspaces
👉 Generates insights

6. 🔗 Dependencies

Receives Data From:

- Academic Workspace
- Student Workspace
- Examination Workspace
- Payment Workspace
- HRMS Workspace
- Timetable Workspace

Sends Actions To:

- All workspaces (via drill-down)

7. 📊 Key Metrics

- Academic performance index
- Attendance percentage
- Fee collection rate
- Operational efficiency

8. 🖥️ UI / UX Structure (Recommended)

Layout:

Top Section (KPIs):

- Attendance | Fees | Results | Activities

Middle Section (Charts):

- Performance graphs
- Trends

Bottom Section:

- Alerts
- Reports

Widgets:

- Cards
- Graphs
- Tables
- Alerts panel

9. ⚠️ Critical Design Notes

- Must be **fast and real-time**
- Should avoid **data duplication**
- Needs **clean and executive-friendly UI**
- Must support **drill-down navigation**

10. 🚀 Why This Workspace is Critical

Without this:

- No centralized decision-making
- No performance insights
- No monitoring

👉 This workspace is the **brain of the institution**

WORKSPACE 14: TRANSPORT COORDINATOR WORKSPACE

1. 🧭 Overview

The **Transport Coordinator Workspace** manages the **entire school transportation system**.

It handles:

- Bus operations
- Route planning
- Student transport allocation
- Tracking and monitoring

This workspace ensures:
👉 Safe student transportation
👉 Efficient route management
👉 Operational transparency

It acts as:

- A **transport operations system**
- A **logistics management module**
- A **student safety support system**

2. 👥 Primary Users

- Transport Coordinator
- Admin / School Management
- Drivers & Attendants (limited access, optional)
- Parents (view access via Parent Portal)

3. 🧩 Complete Module Breakdown

3.1 Transport Dashboard

Central overview of transport operations.

📊 **Displays:**

- Total buses
- Active routes
- Students using transport
- Daily trip status

⚙️ **Features:**

- Alerts (delays, issues)
- Route performance insights

3.2 Bus Management

Manages **vehicle data**.

📌 **Data Includes:**

- Bus number
- Driver details
- Capacity
- Vehicle status

⚙️ **Features:**

- Add/edit buses
- Assign drivers
- Track availability

3.3 Route Management

Handles **route creation and planning**.

⚙️ **Features:**

- Create routes
- Define route paths
- Assign stops

📊 **Example:**

Route A → Stop 1 → Stop 2 → School

3.4 Boarding Points (Stops Management)

Manages **pickup/drop locations**.

⚙️ **Features:**

- Add stops
- Assign stops to routes
- Define timings

3.5 Route-Bus Assignment

Links buses to routes.

⚙️ **Features:**

- Assign bus to route
- Manage route schedules
- Handle multiple shifts

3.6 Student Transport Allocation

Assigns students to transport.

⚙️ **Features:**

- Assign student → route + stop
- Update transport details
- Bulk allocation

3.7 Transport Registration

Handles student transport enrollment.

⚙️ **Features:**

- Register students for transport
- Manage approvals
- Track active/inactive users

3.8 Transport Analytics

Provides operational insights.

📊 **Metrics:**

- Route efficiency
- Bus occupancy
- Usage trends

3.9 Trip Tracking (Advanced / Optional)

Real-time tracking (if GPS enabled).

⚙️ **Features:**

- Live bus location
- Estimated arrival time
- Trip logs

3.10 Notifications & Alerts

Keeps stakeholders informed.

⚙️ **Alerts:**

- Bus delays
- Route changes
- Emergency alerts

3.11 Reports Module

Provides detailed reports.

📊 **Reports:**

- Student transport list
- Route-wise allocation
- Bus utilization

3.12 Configuration Module

Controls system settings.

⚙️ **Features:**

- Define transport rules
- Set route constraints
- Configure alerts

3.13 “What’s New” Module

Platform feature updates.

4. 🔄 End-to-End Workflow

1. Add buses
2. Create routes
3. Define stops
4. Assign buses to routes
5. Register students
6. Allocate students to routes
7. Monitor trips
8. Analyze performance

5. 🧠 Data Ownership

This workspace manages:

- Bus data
- Routes & stops
- Student transport allocation
- Trip logs

6. 🔗 Dependencies

Receives Data From:

- Student Workspace (student data)

Sends Data To:

- Parent Portal (transport info)
- Admin Dashboard (analytics)

7. 📊 Key Metrics

- Bus occupancy rate
- Route efficiency
- Transport usage
- On-time performance

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Buses
- Routes
- Stops
- Assignments
- Analytics
- Reports
- Configuration

Main UI:

Map View (Optional):

- Route visualization

Table View:

- Bus | Route | Students

9. ⚠️ Critical Design Notes

- Must prioritize **student safety**
- Should support **real-time tracking (future-ready)**
- Needs **efficient route optimization**
- Must be **simple for non-technical users**

10. 🚀 Why This Workspace is Important

Without this:

- Transport becomes unmanaged
- No visibility for parents
- Inefficient routes

👉 This workspace is the **logistics and safety backbone for student transportation**

WORKSPACE 15: LIBRARY MANAGEMENT WORKSPACE

1. 🧭 Overview

The **Library Management Workspace** is the **digital library system** of the institution.

It manages:

- Book inventory
- Issue/return operations
- Student & faculty borrowing
- Library usage analytics

This workspace ensures:
👉 Organized book management
👉 Efficient borrowing system
👉 Transparency in library operations

It acts as:

- A **digital catalog system**
- A **circulation management system**
- A **resource tracking module**

2. 👥 Primary Users

- Librarian
- Students
- Faculty
- Admin (oversight)

3. 🧩 Complete Module Breakdown

3.1 Library Dashboard

Central overview of library activity.

📊 **Displays:**

- Total books
- Issued books
- Available books
- Overdue items

⚙️ **Features:**

- Alerts (overdue returns)
- Usage insights

3.2 Book Inventory Management (Core Module)

Maintains the **complete book database**.

📌 **Data Includes:**

- Book title
- Author
- ISBN
- Category/subject
- Number of copies

⚙️ **Features:**

- Add/edit books
- Categorize books
- Track availability

3.3 Catalog & Search System

Provides easy book discovery.

⚙️ **Features:**

- Search by title, author, subject
- Filter by category
- Availability status

3.4 Issue & Return Management

Handles borrowing operations.

⚙️ **Features:**

a) Book Issue

- Assign book to student/faculty
- Set due date

b) Book Return

- Mark returned books
- Update inventory

c) Renewal

- Extend due date

3.5 Fine & Penalty Management

Handles overdue fines.

⚙️ **Features:**

- Calculate fines
- Track penalties
- Payment tracking (optional integration)

3.6 Member Management

Tracks library users.

⚙️ **Features:**

- Student membership
- Faculty membership
- Borrowing limits

3.7 Reservation System

Allows advance booking.

⚙️ **Features:**

- Reserve unavailable books
- Queue system

3.8 Digital Library (Advanced / Optional)

Supports e-learning resources.

⚙️ **Features:**

- Upload PDFs
- Access digital books
- Online reading

3.9 Reports & Analytics

Provides library insights.

📊 **Reports:**

- Issued books report
- Overdue report
- Popular books
- User activity

3.10 Notifications & Alerts

Keeps users informed.

⚙️ **Alerts:**

- Due date reminders
- Overdue alerts
- Reservation notifications

3.11 Configuration Module

Controls system settings.

⚙️ **Features:**

- Borrowing rules
- Fine rules
- Membership limits

3.12 “What’s New” Module

Platform feature updates.

4. 🔄 End-to-End Workflow

1. Add books to inventory
2. Register members
3. Issue books
4. Track due dates
5. Receive returns
6. Apply fines (if needed)
7. Generate reports

5. 🧠 Data Ownership

This workspace manages:

- Book inventory
- Issue/return logs
- Member records
- Fine records

6. 🔗 Dependencies

Receives Data From:

- Student Workspace
- Faculty Workspace

Sends Data To:

- Admin Dashboard (analytics)
- Common Workspace (notifications)

7. 📊 Key Metrics

- Book circulation rate
- Overdue percentage
- Popular books
- Library usage

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Books
- Catalog
- Issue/Return
- Members
- Reports
- Configuration

Main UI:

Table View:

- Book | Author | Status

Search Bar (Important):

- Quick book search

9. ⚠️ Critical Design Notes

- Must support **fast search & filtering**
- Needs **accurate inventory tracking**
- Should handle **multiple copies efficiently**
- Must be **simple for librarian use**

10. 🚀 Why This Workspace is Important

Without this:

- Library operations become manual
- Book tracking is inefficient
- Resource usage is unclear

👉 This workspace is the **knowledge resource management system of the institution**

WORKSPACE 16: PARENT PORTAL WORKSPACE

1. 🧭 Overview

The **Parent Portal Workspace** is the **primary interaction interface for parents/guardians** within the VID system.

It provides:

- Real-time visibility into student progress
- Direct communication with the institution
- Access to financial, academic, and attendance data

This workspace ensures:
👉 Transparency between school and parents
👉 Improved parent engagement
👉 Better student monitoring

It acts as:

- A **monitoring dashboard for parents**
- A **communication bridge**
- A **self-service portal**

2. 👥 Primary Users

- Parents / Guardians
- Class Teachers (interaction)
- Admin (monitoring communication)

3. 🧩 Complete Module Breakdown

3.1 Parent Dashboard

Central overview of student data.

📊 **Displays:**

- Student summary (attendance, performance)
- Fee status
- Notifications
- Upcoming events

⚙️ **Features:**

- Quick access to all modules
- Personalized view per child

3.2 Student Profile View

Detailed student information.

📌 **Includes:**

- Personal details
- Academic details

- Class & section

⚙️ **Features:**

- View-only access
- Multi-child support (if parent has multiple children)

3.3 Attendance Monitoring

Tracks student attendance.

⚙️ **Features:**

- Daily attendance view
- Monthly summary
- Absence alerts

3.4 Academic Performance Module

Provides academic insights.

⚙️ **Features:**

- View marks & grades
- Download report cards
- Subject-wise performance

3.5 Fee & Payment Module

Handles financial visibility.

⚙️ **Features:**

- View fee structure
- Check dues
- Make online payments
- Download receipts

3.6 Timetable Access

Displays student schedule.

⚙️ **Features:**

- Class timetable
- Exam timetable
- Substitution updates

3.7 Communication Module

Enables parent-school communication.

⚙️ **Features:**

- Message teachers
- Receive announcements
- Participate in discussions

3.8 Notice Board

Displays official announcements.

⚙️ **Features:**

- View school notices
- Download attachments
- Acknowledge notices

3.9 Event & Activity Tracking

Tracks student participation.

⚙️ **Features:**

- View upcoming events
- Participation details
- Achievements

3.10 Transport Tracking (If Enabled)

Provides transport visibility.

⚙️ **Features:**

- Bus route details
- Pickup/drop timings
- Live tracking (optional)

3.11 Health Monitoring (If Enabled)

Tracks student health records.

⚙️ **Features:**

- Medical records
- Alerts
- Emergency updates

3.12 Requests & Permissions

Handles parent requests.

⚙️ **Features:**

- Leave requests
- Permission requests
- Meeting requests

3.13 Notifications & Alerts

Keeps parents updated.

⚙️ **Alerts:**

- Attendance alerts
- Fee reminders

- Exam updates
- Emergency alerts

3.14 Configuration & Profile Settings

Manages parent account.

⚙️ **Features:**

- Update profile
- Notification preferences
- Password/security settings

3.15 "What's New" Module

Displays new features/updates.

4. 🔄 End-to-End Workflow

1. Parent logs in
2. Views dashboard summary
3. Checks attendance & performance
4. Pays fees
5. Communicates with school
6. Receives alerts and updates

5. 🧠 Data Ownership

This workspace:

- Does not own core data

It:
👉 Aggregates student-related data
👉 Provides **read-only + interaction access**

6. 🔗 Dependencies

Receives Data From:

- Student Workspace
- Examination Workspace
- Payment Workspace
- Timetable Workspace
- Transport Workspace
- Health Workspace

7. 📊 Key Metrics

- Parent engagement rate
- Payment completion rate
- Communication activity
- Attendance monitoring frequency

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Student Profile
- Attendance
- Academics
- Fees
- Timetable
- Communication
- Requests
- Settings

Main UI:

Card-Based Design:

- Attendance card
- Fee card
- Performance card

👉 Must be **mobile-first & highly responsive**

9. ⚠️ Critical Design Notes

- Must be **simple and intuitive**
- Should be **mobile optimized (very important)**
- Needs **real-time updates**
- Must ensure **data privacy (parent-specific data)**

10. 🚀 Why This Workspace is Critical

Without this:

- Parents have no visibility
- Communication gap increases
- Engagement drops

👉 This workspace is the **parent engagement engine of the institution**

WORKSPACE 17: EVENT MANAGEMENT WORKSPACE

1. 🧭 Overview

The **Event Management Workspace** handles the **planning, execution, and tracking of all institutional events and activities**.

This includes:

- Academic events (exams, seminars)
- Co-curricular activities (competitions, clubs)
- Cultural events (annual day, festivals)
- Administrative events (meetings, orientations)

This workspace ensures:
👉 Structured event planning
👉 Efficient participation management
👉 Visibility across stakeholders

It acts as:

- An **event planning system**
- A **participation tracking platform**
- A **calendar coordination tool**

2. 👥 Primary Users

- Event Coordinator
- Faculty
- Admin / Principal
- Students (participants)
- Parents (view access)

3. 🧩 Complete Module Breakdown

3.1 Event Dashboard

Central overview of all events.

📊 **Displays:**

- Upcoming events
- Ongoing events
- Completed events
- Participation stats

⚙️ **Features:**

- Alerts for upcoming deadlines
- Event highlights

3.2 Event Creation & Planning (Core Module)

Handles event setup.

📌 **Data Includes:**

- Event name
- Event type
- Date & time
- Venue
- Organizer

⚙️ **Features:**

- Create/edit events
- Define event agenda
- Set participation rules

3.3 Event Calendar

Provides visual scheduling.

⚙️ **Features:**

- Monthly/weekly calendar view
- Event timeline
- Conflict checking

3.4 Participation Management

Handles registrations.

⚙️ **Features:**

- Student registration
- Participant lists
- Capacity control

3.5 Event Roles & Assignments

Manages responsibilities.

⚙️ **Features:**

- Assign coordinators
- Assign volunteers
- Task allocation

3.6 Resource & Venue Management

Handles event logistics.

⚙️ **Features:**

- Allocate rooms/halls
- Manage equipment
- Track availability

3.7 Attendance Tracking (Event-Level)

Tracks event participation.

⚙️ **Features:**

- Mark attendance
- Record participation
- Generate participation logs

3.8 Results & Achievements

Stores outcomes.

⚙️ **Features:**

- Record winners
- Assign certificates
- Track achievements

3.9 Communication Module

Handles event communication.

⚙️ **Features:**

- Send invitations
- Broadcast updates
- Notify participants

3.10 Reports & Analytics

Provides insights.

📊 **Reports:**

- Event participation report
- Attendance report
- Event performance

3.11 Notifications & Alerts

Keeps stakeholders informed.

⚙️ **Alerts:**

- Event reminders
- Registration deadlines
- Schedule updates

3.12 Configuration Module

Controls event settings.

⚙️ **Features:**

- Define event types
- Set participation rules
- Configure notifications

3.13 “What’s New” Module

Displays new feature updates.

4. 🔄 End-to-End Workflow

1. Create event
2. Define details & schedule
3. Open registrations
4. Assign roles
5. Conduct event
6. Mark attendance
7. Record results
8. Generate reports

5. 🧠 Data Ownership

This workspace manages:

- Event data
- Participation records
- Attendance logs
- Results & achievements

6. 🔗 Dependencies

Receives Data From:

- Student Workspace
- Faculty Workspace

Sends Data To:

- Parent Portal (event visibility)
- Admin Dashboard (analytics)
- Common Workspace (notifications)

7. 📊 Key Metrics

- Event participation rate
- Attendance rate
- Event success metrics
- Engagement levels

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Events
- Calendar
- Participants
- Reports
- Configuration

Main UI:

Calendar View (Primary UI):

- Visual event planning

Event Cards:

- Quick event summary

9. ⚠️ Critical Design Notes

- Must support **multiple event types**
- Needs **calendar integration**
- Should be **easy to use for coordinators**
- Must allow **real-time updates**

10. 🚀 Why This Workspace is Important

Without this:

- Events become unorganized
- Participation tracking is difficult
- No engagement visibility

👉 This workspace is the **activity and engagement management system of the institution**

WORKSPACE 18: DISCIPLINARY COMMITTEE WORKSPACE

1. 🧭 Overview

The **Disciplinary Committee Workspace** is responsible for managing and monitoring **student behavior, discipline, and institutional compliance**.

It handles:

- Incident reporting
- Disciplinary actions
- Behavior tracking
- Escalations & resolutions

This workspace ensures:
👉 Student discipline is systematically managed
👉 Transparency in disciplinary actions
👉 Fair and auditable processes

It acts as:

- A **behavior tracking system**
- A **compliance and governance module**
- A **case management system**

2. 👥 Primary Users

- Disciplinary Committee Members
- Class Teachers
- Principal / Admin (oversight)
- Counselors (if available)

3. 🧩 Complete Module Breakdown

3.1 Discipline Dashboard

Central overview of disciplinary activity.

📊 **Displays:**

- Total incidents
- Pending cases
- Resolved cases
- High-risk students

⚙️ **Features:**

- Alerts for serious incidents
- Case summaries

3.2 Incident Reporting (Core Module)

Captures all disciplinary issues.

📌 **Data Includes:**

- Student details
- Date & time
- Incident type
- Description
- Evidence (images/docs)

⚙️ **Features:**

- Report incident
- Attach proof
- Categorize severity (low/medium/high)

3.3 Case Management System

Handles full lifecycle of incidents.

⚙️ **Features:**

- Create case from incident
- Assign committee members
- Track case status

📊 **Case Status:**

- Open
- Under Review
- Resolved
- Escalated

3.4 Action & Resolution Tracking

Tracks disciplinary actions.

⚙️ **Features:**

- Record actions (warning, suspension, counseling)
- Add remarks
- Track follow-ups

3.5 Student Behavior Profile

Maintains behavioral history.

⚙️ **Features:**

- View all past incidents
- Track behavior patterns
- Risk assessment

3.6 Escalation Management

Handles serious cases.

⚙️ **Features:**

- Escalate to higher authority
- Notify principal

- Special review workflows

3.7 Parent Communication

Informs parents about incidents.

⚙️ **Features:**

- Send notifications
- Schedule meetings
- Share reports

3.8 Counseling & Rehabilitation Tracking

Supports corrective actions.

⚙️ **Features:**

- Assign counseling sessions
- Track progress
- Record outcomes

3.9 Reports & Analytics

Provides discipline insights.

📊 **Reports:**

- Incident reports
- Student behavior trends
- Case resolution rates

3.10 Notifications & Alerts

Keeps stakeholders informed.

⚙️ **Alerts:**

- New incidents
- Escalations
- Pending cases

3.11 Audit Trail & Compliance

Ensures transparency.

⚙️ **Features:**

- Track all actions
- Maintain logs
- Ensure accountability

3.12 Configuration Module

Controls discipline rules.

⚙️ **Features:**

- Define incident categories
- Set severity levels
- Configure workflows

3.13 “What’s New” Module

Displays updates/features.

4. 🔄 End-to-End Workflow

1. Incident is reported
2. Case is created
3. Committee reviews case
4. Action is assigned
5. Parent is notified
6. Case is resolved
7. Behavior is tracked

5. 🧠 Data Ownership

This workspace manages:

- Incident records
- Case files
- Behavior profiles
- Action logs

6. 🔗 Dependencies

Receives Data From:

- Student Workspace
- Faculty Workspace

Sends Data To:

- Parent Portal (notifications)
- Admin Dashboard (analytics)

7. 📊 Key Metrics

- Number of incidents
- Resolution time
- Repeat offenders
- Behavior improvement rate

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Incidents
- Cases
- Students
- Reports
- Configuration

Main UI:

Case Table:

- Student | Incident | Status | Severity

Detail View:

- Full case history

9. ⚠️ Critical Design Notes

- Must ensure **data privacy (sensitive data)**
- Needs **audit trail for accountability**
- Should support **escalation workflows**
- Must be **structured and transparent**

10. 🚀 Why This Workspace is Important

Without this:

- No structured discipline management
- No behavior tracking
- No accountability

👉 This workspace is the **governance and compliance backbone of student behavior management**

WORKSPACE 19: SPORTS OFFICER WORKSPACE

1. 🧭 Overview

The **Sports Officer Workspace** manages all **sports activities, physical education programs, and athletic performance tracking** within the institution.

It handles:

- Sports events
- Team management
- Student participation
- Performance tracking

This workspace ensures:
👉 Structured sports management
👉 Talent identification
👉 Performance monitoring

It acts as:

- A **sports operations system**
- A **team management platform**
- A **performance tracking system**

2. 👥 Primary Users

- Sports Officer
- Coaches
- Students (participants)
- Admin / Principal (oversight)

3. 🧩 Complete Module Breakdown

3.1 Sports Dashboard

Central overview of sports activities.

📊 **Displays:**

- Active sports events
- Teams and players
- Upcoming matches
- Performance highlights

⚙️ **Features:**

- Alerts (events, matches)
- Participation insights

3.2 Sports Event Management

Handles sports-specific events.

⚙️ **Features:**

- Create sports events
- Define match schedules
- Manage tournaments

3.3 Team Management (Core Module)

Manages sports teams.

📌 **Data Includes:**

- Team name
- Sport type (Cricket, Football, etc.)
- Players list
- Coach details

⚙️ **Features:**

- Create teams
- Add/remove players
- Assign coaches

3.4 Player Management

Tracks individual student athletes.

⚙️ **Features:**

- Player profiles
- Sports participation
- Skill tracking

3.5 Match & Tournament Scheduling

Manages competitions.

⚙️ **Features:**

- Schedule matches
- Create tournament brackets
- Assign teams

3.6 Performance Tracking

Tracks athletic performance.

⚙️ **Features:**

- Match results
- Player statistics
- Team performance

3.7 Attendance & Participation Tracking

Tracks sports involvement.

⚙️ **Features:**

- Mark attendance for practice
- Track event participation

3.8 Achievements & Certificates

Records accomplishments.

⚙️ **Features:**

- Record awards
- Generate certificates
- Track achievements

3.9 Resource & Equipment Management

Handles sports resources.

⚙️ **Features:**

- Track equipment
- Manage inventory
- Allocate resources

3.10 Communication Module

Handles sports-related communication.

⚙️ **Features:**

- Notify players
- Share schedules
- Send updates

3.11 Reports & Analytics

Provides insights.

📊 **Reports:**

- Team performance
- Player stats
- Event reports

3.12 Notifications & Alerts

Keeps stakeholders informed.

⚙️ **Alerts:**

- Match reminders
- Practice schedules
- Result updates

3.13 Configuration Module

Controls sports settings.

⚙️ **Features:**

- Define sports categories
- Set team rules
- Configure scoring formats

3.14 “What’s New” Module

Displays updates/features.

4. 🔄 End-to-End Workflow

1. Create sports event
2. Form teams
3. Register players
4. Schedule matches
5. Conduct events
6. Track performance
7. Record achievements

5. 🧠 Data Ownership

This workspace manages:

- Teams and players
- Match schedules
- Performance data
- Achievements

6. 🔗 Dependencies

Receives Data From:

- Student Workspace

Sends Data To:

- Parent Portal (achievements visibility)
- Admin Dashboard (performance insights)

7. 📊 Key Metrics

- Participation rate
- Team performance
- Player achievements
- Event engagement

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Events
- Teams
- Players
- Matches
- Reports
- Configuration

Main UI:

Team Table:

- Team | Sport | Players

Match Schedule View:

- Date | Teams | Venue

9. ⚠️ Critical Design Notes

- Must support **multiple sports types**
- Should be **easy to manage teams**
- Needs **performance tracking clarity**
- Must be **engaging and visual**

10. 🚀 Why This Workspace is Important

Without this:

- Sports activities become unstructured
- No tracking of performance
- No recognition of talent

👉 This workspace is the **sports and physical development management system of the institution**

WORKSPACE 20: HEALTH MONITORING WORKSPACE

1. 🧭 Overview

The **Health Monitoring Workspace** manages the **medical records, health tracking, and emergency handling** of students within the institution.

It ensures:

- Continuous monitoring of student health
- Quick response to medical emergencies
- Proper record-keeping for compliance

This workspace is especially important for:
👉 Student safety
👉 Medical history tracking
👉 Emergency preparedness

It acts as:

- A **student health record system**
- An **incident and emergency management module**
- A **preventive health monitoring tool**

2. 👥 Primary Users

- School Nurse / Health Coordinator
- Admin / Principal
- Teachers (limited access)
- Parents (view access via portal)

3. 🧩 Complete Module Breakdown

3.1 Health Dashboard

Central overview of student health data.

📊 **Displays:**

- Students with medical conditions
- Recent health incidents
- Emergency alerts
- Health trends

⚙️ **Features:**

- Quick alerts for critical cases
- Health summaries

3.2 Student Health Records (Core Module)

Maintains complete medical profiles.

📌 **Data Includes:**

- Blood group

- Allergies
- Chronic conditions
- Vaccination records

⚙️ **Features:**

- Add/update medical data
- Upload medical documents
- Track health history

3.3 Medical Incident Management

Handles health-related incidents.

⚙️ **Features:**

- Log incidents (injury, illness)
- Record treatment provided
- Track recovery

3.4 Emergency Response Management

Supports urgent situations.

⚙️ **Features:**

- Emergency alerts
- Contact parents immediately
- Record emergency actions

3.5 Medication & Treatment Tracking

Tracks ongoing treatments.

⚙️ **Features:**

- Record prescribed medicines
- Track dosage schedules
- Monitor treatment progress

3.6 Health Checkup Management

Handles routine health checkups.

⚙️ **Features:**

- Schedule checkups
- Record results
- Track periodic health data

3.7 Alerts & Risk Identification

Identifies health risks.

⚙️ **Features:**

- Flag high-risk students

- Generate alerts for allergies
- Notify relevant staff

3.8 Parent Communication

Keeps parents informed.

⚙️ **Features:**

- Share health updates
- Notify incidents
- Provide medical reports

3.9 Reports & Analytics

Provides health insights.

📊 **Reports:**

- Health records report
- Incident report
- Risk analysis

3.10 Document Management

Stores medical documents.

⚙️ **Features:**

- Upload prescriptions
- Store reports
- Secure access

3.11 Notifications & Alerts

Real-time communication.

⚙️ **Alerts:**

- Emergency alerts
- Medication reminders
- Health warnings

3.12 Configuration Module

Controls system settings.

⚙️ **Features:**

- Define health categories
- Configure alerts
- Set medical workflows

3.13 “What’s New” Module

Displays feature updates.

4. 🔄 End-to-End Workflow

1. Add student health records
2. Track medical history
3. Log incidents
4. Handle emergencies
5. Notify parents
6. Monitor recovery
7. Generate reports

5. 🧠 Data Ownership

This workspace manages:

- Student health records
- Medical incidents
- Treatment logs
- Emergency records

6. 🔗 Dependencies

Receives Data From:

- Student Workspace

Sends Data To:

- Parent Portal (health updates)
- Admin Dashboard (analytics)

7. 📊 Key Metrics

- Number of incidents
- Response time
- Health risk cases
- Recovery rates

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Health Records
- Incidents
- Checkups
- Reports
- Configuration

Main UI:

Health Profile View:

- Student | Conditions | Alerts

Incident Log Table:

- Date | Student | Issue | Status

9. ⚠️ Critical Design Notes

- Must ensure **high data privacy (medical data)**
- Needs **fast emergency response system**
- Should support **alerts and notifications**
- Must be **secure and compliant**

10. 🚀 Why This Workspace is Critical

Without this:

- No health tracking
- No emergency readiness
- No medical history

👉 This workspace is the **student safety and health backbone of the institution**

WORKSPACE 21: HOSTEL ADMINISTRATION WORKSPACE

1. 🧭 Overview

The **Hostel Administration Workspace** manages all **hostel-related operations**, including accommodation, attendance, movement control, and resident services.

It is applicable for:

- Residential schools
- Boarding institutions

This workspace ensures:
👉 Safe accommodation management
👉 Student movement tracking
👉 Hostel discipline and operations

It acts as:

- A **hostel allocation system**
- A **residential monitoring platform**
- A **facility management module**

2. 👥 Primary Users

- Hostel Warden
- Admin
- Students (hostellers)
- Parents (limited visibility)

3. 🧩 Complete Module Breakdown

3.1 Hostel Dashboard

Central overview of hostel operations.

📊 **Displays:**

- Total hostellers
- Occupied rooms
- Vacant rooms
- Pending requests

⚙️ **Features:**

- Alerts for issues
- Quick summaries

3.2 Hostel Allocation (Core Module)

Manages room assignments.

📌 **Data Includes:**

- Hostel building
- Room number
- Bed allocation
- Student assignment

⚙️ **Features:**

- Allocate rooms
- Change room assignments
- Track occupancy

3.3 Room & Infrastructure Management

Handles physical hostel structure.

⚙️ **Features:**

- Manage buildings
- Define rooms & beds
- Track capacity

3.4 Hostel Attendance Tracking

Tracks resident presence.

⚙️ **Features:**

- Daily attendance (morning/night)
- Absentee tracking
- Attendance logs

3.5 Gate Pass Management

Controls student movement.

⚙️ **Features:**

- Request gate pass
- Approve/reject requests
- Track entry/exit logs

3.6 Visitor Management

Handles external visits.

⚙️ **Features:**

- Register visitors
- Track visit logs
- Ensure security

3.7 Complaint & Maintenance Management

Handles hostel issues.

⚙️ **Features:**

- Raise complaints
- Track maintenance requests
- Resolve issues

3.8 Mess & Food Management (Optional but Recommended)

Manages food services.

⚙️ **Features:**

- Meal planning
- Menu scheduling
- Feedback collection

3.9 Student Hostel Profile

Maintains hosteller records.

⚙️ **Features:**

- Room details
- Attendance history
- Gate pass logs

3.10 Reports & Analytics

Provides operational insights.

📊 **Reports:**

- Occupancy report
- Attendance report
- Gate pass report

3.11 Notifications & Alerts

Keeps stakeholders informed.

⚙️ **Alerts:**

- Gate pass approvals
- Attendance alerts
- Complaint updates

3.12 Security & Monitoring

Ensures safety.

⚙️ **Features:**

- Track movements
- Monitor access
- Maintain logs

3.13 Configuration Module

Controls hostel settings.

⚙️ **Features:**

- Define hostel rules
- Configure attendance timings
- Set approval workflows

3.14 “What’s New” Module

Displays updates/features.

4. 🔄 End-to-End Workflow

1. Define hostel structure
2. Allocate rooms to students
3. Track attendance daily
4. Manage gate passes
5. Handle complaints
6. Monitor activities
7. Generate reports

5. 🧠 Data Ownership

This workspace manages:

- Room allocations
- Attendance logs
- Gate pass records
- Complaint logs

6. 🔗 Dependencies

Receives Data From:

- Student Workspace

Sends Data To:

- Parent Portal (hostel updates)
- Admin Dashboard (analytics)

7. 📊 Key Metrics

- Occupancy rate
- Attendance compliance
- Gate pass frequency
- Complaint resolution time

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Hostellers
- Rooms
- Attendance
- Gate Pass

- Reports
- Configuration

Main UI:

Room Allocation Table:

- Room | Bed | Student

Gate Pass Table:

- Student | Date | Status

9. ⚠️ Critical Design Notes

- Must ensure **student safety and tracking**
- Needs **strict access control**
- Should support **real-time monitoring**
- Must be **secure and auditable**

10. 🚀 Why This Workspace is Important

Without this:

- No control over hostel operations
- No tracking of student movement
- No centralized accommodation system

👉 This workspace is the **residential operations and safety management system**

WORKSPACE 22: ALUMNI MANAGEMENT WORKSPACE

1. 🧭 Overview

The **Alumni Management Workspace** manages the **relationship between the institution and its former students (alumni)**.

It focuses on:

- Alumni database management
- Engagement and communication
- Networking opportunities
- Institutional growth through alumni

This workspace ensures:
👉 Long-term connection with past students
👉 Alumni-driven opportunities (mentorship, support)
👉 Strong institutional reputation

It acts as:

- An **alumni database system**
- A **communication and engagement platform**
- A **networking ecosystem**

2. 👥 Primary Users

- Alumni Coordinator
- Admin / Management
- Alumni (external users)
- Current students (limited interaction)

3. 🧩 Complete Module Breakdown

3.1 Alumni Dashboard

Central overview of alumni data.

📊 **Displays:**

- Total alumni count
- Active alumni
- Recent activities
- Engagement levels

⚙️ **Features:**

- Quick insights
- Alerts for updates

3.2 Alumni Database (Core Module)

Stores alumni records.

📌 **Data Includes:**

- Name
- Batch / Year
- Course / Class
- Contact details
- Current profession

⚙️ **Features:**

- Add/edit alumni
- Import alumni data
- Search and filter

3.3 Alumni Profile Management

Detailed alumni profiles.

⚙️ **Features:**

- Professional details
- Achievements
- Social links (optional)

3.4 Engagement & Communication

Handles alumni interaction.

⚙️ **Features:**

- Send newsletters
- Broadcast updates
- Email/SMS communication

3.5 Events & Reunions

Manages alumni events.

⚙️ **Features:**

- Organize reunions
- Track participation
- Send invitations

3.6 Networking & Mentorship

Builds connections.

⚙️ **Features:**

- Alumni-student mentorship
- Networking groups
- Career guidance

3.7 Contributions & Donations (Optional)

Tracks alumni support.

⚙️ **Features:**

- Record donations
- Manage contributions
- Track sponsorships

3.8 Job & Opportunity Sharing

Supports career growth.

⚙️ **Features:**

- Share job openings
- Internship opportunities
- Career referrals

3.9 Reports & Analytics

Provides engagement insights.

📊 **Reports:**

- Alumni engagement report
- Event participation
- Contribution analysis

3.10 Notifications & Alerts

Keeps alumni informed.

⚙️ **Alerts:**

- Event invitations
- News updates
- Important announcements

3.11 Document & Media Management

Stores alumni-related media.

⚙️ **Features:**

- Upload photos
- Store event media
- Archive newsletters

3.12 Configuration Module

Controls alumni system settings.

⚙️ **Features:**

- Define alumni categories
- Configure communication rules
- Set engagement workflows

3.13 “What’s New” Module

Displays updates/features.

4. 🔄 End-to-End Workflow

1. Add alumni records
2. Maintain profiles
3. Engage via communication
4. Organize events
5. Enable networking
6. Track contributions
7. Generate reports

5. 🧠 Data Ownership

This workspace manages:

- Alumni database
- Engagement records
- Event participation
- Contributions

6. 🔗 Dependencies

Receives Data From:

- Student Workspace (graduated students)

Sends Data To:

- Admin Dashboard (analytics)
- Event Workspace (alumni events)

7. 📊 Key Metrics

- Alumni engagement rate
- Event participation
- Contribution value
- Active alumni percentage

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Alumni
- Events
- Communication
- Opportunities
- Reports
- Configuration

Main UI:

Alumni Table:

- Name | Batch | Profession

Engagement Panel:

- Messages | Events | Updates

9. ⚠️ Critical Design Notes

- Must support **long-term data storage**
- Needs **strong communication tools**
- Should enable **networking features**
- Must be **scalable for large alumni base**

10. 🚀 Why This Workspace is Important

Without this:

- Alumni connections are lost
- No long-term engagement
- No external network growth

👉 This workspace is the **institution's lifelong relationship and networking system**

WORKSPACE 23: PLACEMENT CELL WORKSPACE

1. 🧭 Overview

The **Placement Cell Workspace** manages **career opportunities, job placements, and student career tracking**.

⚠️ Note: While this is more critical for colleges, in a **school context**, it can be adapted for:

- Career guidance (higher studies)
- Internship opportunities
- Skill development tracking

This workspace ensures:
👉 Structured career support
👉 Opportunity tracking
👉 Student outcome visibility

It acts as:

- A **career management system**
- An **opportunity tracking platform**
- A **student outcome monitoring tool**

2. 👥 Primary Users

- Placement Officer / Career Counselor
- Students (higher classes)
- Admin / Principal
- External Organizations (optional)

3. 🧩 Complete Module Breakdown

3.1 Placement Dashboard

Central overview of placement activities.

📊 **Displays:**

- Total opportunities
- Students placed / selected
- Active applications
- Upcoming drives

⚙️ **Features:**

- Career insights
- Opportunity alerts

3.2 Opportunity Management (Core Module)

Manages career opportunities.

📌 **Data Includes:**

- Opportunity type (job/internship/course)
- Organization name
- Eligibility criteria
- Application deadlines

⚙️ **Features:**

- Create opportunities
- Edit details
- Publish opportunities

3.3 Student Registration & Eligibility

Tracks student participation.

⚙️ **Features:**

- Student registration
- Eligibility filtering
- Application tracking

3.4 Application Management

Handles applications lifecycle.

⚙️ **Features:**

- Apply for opportunities
- Track application status
- Shortlisting

📊 **Status:**

- Applied
- Shortlisted
- Selected
- Rejected

3.5 Organization / Partner Management

Manages external entities.

⚙️ **Features:**

- Maintain company database
- Track partnerships
- Manage contacts

3.6 Career Guidance & Counseling

Supports student development.

⚙️ **Features:**

- Career sessions
- Guidance resources
- Counseling records

3.7 Interview & Selection Process

Manages selection stages.

⚙️ **Features:**

- Schedule interviews
- Track rounds
- Record results

3.8 Student Career Profile

Maintains career data.

⚙️ **Features:**

- Skills and interests
- Achievements
- Career preferences

3.9 Reports & Analytics

Provides placement insights.

📊 **Reports:**

- Placement report
- Application success rate
- Opportunity analysis

3.10 Notifications & Alerts

Keeps students informed.

⚙️ **Alerts:**

- New opportunities
- Application deadlines
- Selection updates

3.11 Document Management

Handles career documents.

⚙️ **Features:**

- Upload resumes
- Store certificates
- Manage documents

3.12 Configuration Module

Controls system settings.

⚙️ **Features:**

- Define opportunity types
- Configure eligibility rules

- Set workflows

3.13 “What’s New” Module

Displays updates/features.

4. 🔄 End-to-End Workflow

1. Create opportunity
2. Define eligibility
3. Students apply
4. Shortlisting process
5. Conduct interviews
6. Final selection
7. Track outcomes

5. 🧠 Data Ownership

This workspace manages:

- Opportunities
- Applications
- Student career data
- Organization records

6. 🔗 Dependencies

Receives Data From:

- Student Workspace
- Academic Workspace (performance data)

Sends Data To:

- Admin Dashboard (analytics)
- Parent Portal (optional visibility)

7. 📊 Key Metrics

- Placement rate
- Application success rate
- Student participation
- Opportunity count

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Opportunities
- Applications
- Organizations
- Reports
- Configuration

Main UI:

Opportunity Table:

- Name | Type | Deadline

Application Panel:

- Student | Status | Result

9. ⚠️ Critical Design Notes

- Must support **school-level adaptation (career guidance)**
- Needs **clear application tracking**
- Should allow **external integrations (optional)**
- Must be **student-friendly**

10. 🚀 Why This Workspace is Important

Without this:

- No structured career guidance
- No opportunity tracking
- No student outcome visibility

👉 This workspace is the **career transition and opportunity management system**

WORKSPACE 24: RESEARCH & DEVELOPMENT (R&D) WORKSPACE

1. 🧭 Overview

The **Research & Development (R&D) Workspace** manages **innovation, projects, academic exploration, and knowledge development** within the institution.

⚠️ Important Adaptation for Schools:
In schools, this is not traditional "research" like colleges. Instead, it focuses on:

- Student projects
- Science exhibitions
- Innovation activities
- Skill-based learning

This workspace ensures:
👉 Encouragement of creativity and innovation
👉 Structured project tracking
👉 Academic enrichment beyond textbooks

It acts as:

- An **innovation management system**
- A **project tracking platform**
- A **skill and creativity development module**

2. 👥 Primary Users

- Academic Coordinators
- Teachers (project guides)
- Students
- Admin / Principal

3. 🧩 Complete Module Breakdown

3.1 R&D Dashboard

Central overview of all innovation activities.

📊 **Displays:**

- Active projects
- Completed projects
- Student participation
- Upcoming events (science fairs, exhibitions)

⚙️ **Features:**

- Innovation highlights
- Alerts for submissions

3.2 Project Management (Core Module)

Handles all student/teacher projects.

📌 **Data Includes:**

- Project title
- Category (Science, Tech, Arts, etc.)
- Students involved
- Guide/mentor

⚙️ **Features:**

- Create project
- Assign students
- Track progress

3.3 Innovation & Idea Submission

Encourages idea generation.

⚙️ **Features:**

- Submit ideas
- Review ideas
- Approve/reject concepts

3.4 Project Evaluation & Assessment

Evaluates projects.

⚙️ **Features:**

- Assign evaluators
- Define evaluation criteria
- Score projects

3.5 Exhibitions & Competitions

Manages events like science fairs.

⚙️ **Features:**

- Organize exhibitions
- Register participants
- Track winners

3.6 Student Innovation Profiles

Tracks student creativity.

⚙️ **Features:**

- Maintain project history
- Track achievements
- Record innovation skills

3.7 Collaboration & Mentorship

Supports teamwork.

⚙️ **Features:**

- Assign mentors
- Enable group projects
- Track mentor feedback

3.8 Documentation & Repository

Stores project data.

⚙️ **Features:**

- Upload reports
- Store presentations
- Maintain archives

3.9 Reports & Analytics

Provides insights.

📊 **Reports:**

- Project participation
- Innovation trends
- Student performance

3.10 Notifications & Alerts

Keeps users informed.

⚙️ **Alerts:**

- Submission deadlines
- Evaluation updates
- Event notifications

3.11 Certification & Recognition

Recognizes achievements.

⚙️ **Features:**

- Generate certificates
- Record awards
- Track recognitions

3.12 Configuration Module

Controls system settings.

⚙️ **Features:**

- Define project categories
- Configure evaluation rules
- Set workflows

3.13 “What’s New” Module

Displays updates/features.

4. 🔄 End-to-End Workflow

1. Submit idea
2. Approve project
3. Assign mentor
4. Develop project
5. Evaluate project
6. Showcase in exhibition
7. Record achievements

5. 🧠 Data Ownership

This workspace manages:

- Project data
- Innovation records
- Evaluation scores
- Achievements

6. 🔗 Dependencies

Receives Data From:

- Student Workspace
- Faculty Workspace

Sends Data To:

- Admin Dashboard (analytics)
- Event Workspace (exhibitions)

7. 📊 Key Metrics

- Project participation rate
- Innovation submissions
- Competition performance
- Student engagement

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Projects
- Ideas
- Exhibitions
- Reports
- Configuration

Main UI:

Project Table:

- Project | Students | Status

Idea Board:

- Submitted ideas

9. ⚠️ Critical Design Notes

- Must be **simple and engaging for students**
- Should encourage **creativity and participation**
- Needs **structured evaluation system**
- Must support **file/document uploads**

10. 🚀 Why This Workspace is Important

Without this:

- No structured innovation system
- No project tracking
- Limited student creativity exposure

👉 This workspace is the **innovation and creativity engine of the institution**

WORKSPACE 25: TEAM OWNER (SUPER ADMIN) WORKSPACE

1. 🧭 Overview

The **Team Owner Workspace (Super Admin)** is the **highest authority control layer** of the entire VID platform.

It operates at:
👉 **Global level (across all institutions / tenants)**

This workspace is responsible for:

- Managing institutions
- Controlling subscriptions
- Monitoring system performance
- Governing the entire platform

It ensures:
👉 Centralized governance
👉 Platform scalability
👉 Tenant isolation & control

It acts as:

- A **platform control center**
- A **multi-tenant management system**
- A **system monitoring and governance layer**

2. 👥 Primary Users

- Platform Owner
- Super Admin Team
- System Administrators

3. 🧩 Complete Module Breakdown

3.1 Super Admin Dashboard

Global overview of the platform.

📊 **Displays:**

- Total institutions
- Active users
- System usage metrics
- Revenue / subscriptions

⚙️ **Features:**

- Real-time monitoring
- Alerts for issues

3.2 Institution Management (Core Module)

Handles tenant lifecycle.

📌 **Data Includes:**

- Institution name
- Code / ID
- Address
- Contact details
- Branding (logo, favicon)

⚙️ **Features:**

- Create institution
- Activate / deactivate institution
- Edit institution details
- Assign plans

3.3 Organization Settings

Manages platform-level configurations.

⚙️ **Features:**

- Define global policies
- Configure system defaults
- Manage global parameters

3.4 Subscription & Billing Management

Handles monetization.

⚙️ **Features:**

- Create subscription plans
- Assign plans to institutions
- Track renewals
- Payment tracking

3.5 User Management (Global Level)

Controls platform users.

⚙️ **Features:**

- Manage super admins
- Assign roles
- Control permissions

3.6 App & Workspace Management

Controls system modules.

⚙️ **Features:**

- Enable/disable workspaces per institution
- Manage feature access
- Deploy new modules

3.7 Data Management

Handles global data operations.

⚙️ **Features:**

- Backup & restore
- Data migration
- Data cleanup

3.8 System Monitoring & Logs

Tracks platform health.

⚙️ **Features:**

- Monitor uptime
- Error tracking
- Activity logs

3.9 Security & Access Control

Ensures platform security.

⚙️ **Features:**

- Access control policies
- Audit logs
- Security monitoring

3.10 Reports & Analytics

Provides global insights.

📊 **Reports:**

- Institution usage
- Revenue reports
- Performance metrics

3.11 Notifications & Alerts

System-level alerts.

⚙️ **Alerts:**

- System issues
- Subscription expiry
- Critical events

3.12 API & Integration Management

Handles external integrations.

⚙️ **Features:**

- Manage APIs
- Integrate third-party services
- Monitor API usage

3.13 Configuration Module

Controls system-wide settings.

⚙️ **Features:**

- Global configuration
- Default settings
- Feature toggles

3.14 “What’s New” Module

Displays platform updates.

4. 🔄 End-to-End Workflow

1. Create institution
2. Assign subscription
3. Enable workspaces
4. Monitor usage
5. Manage data
6. Track performance
7. Ensure system health

5. 🧠 Data Ownership

This workspace manages:

- Institution registry
- Subscription data
- Platform analytics
- System logs

6. 🔗 Dependencies

Controls All Workspaces

Sends Data To:

- Admin Dashboards (aggregated insights)

7. 📊 Key Metrics

- Total institutions
- Active users
- Revenue growth
- System uptime

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Institutions
- Subscriptions
- Users

- Apps
- Reports
- Settings

Main UI:

Institution Table:

- Name | Status | Plan

System Monitoring Panel:

- Uptime | Errors | Alerts

9. ⚠️ Critical Design Notes

- Must support **multi-tenant architecture**
- Needs **high security & access control**
- Should handle **large-scale data**
- Must ensure **high availability**

10. 🚀 Why This Workspace is Critical

Without this:

- No platform control
- No tenant management
- No scalability

👉 This workspace is the **brain and control center of the entire VID platform**

WORKSPACE 26: USER MANAGEMENT SYSTEM (UMS) WORKSPACE

1. 🧭 Overview

The **User Management System (UMS) Workspace** is the **identity and access control backbone** of the entire VID platform.

It is responsible for:

- User creation
- Role assignment
- Permission control
- Identity management

This workspace ensures:
👉 Secure access to the system
👉 Proper role-based control (RBAC)
👉 Data privacy and authorization

It acts as:

- An **identity management system**
- An **access control engine**
- A **security enforcement layer**

2. 👥 Primary Users

- Institution Admin
- HR/Admin Team
- Super Admin (global level)

3. 🧩 Complete Module Breakdown

3.1 UMS Dashboard

Central overview of all users.

📊 **Displays:**

- Total users
- Active users
- Role distribution
- Recent logins

⚙️ **Features:**

- Quick insights
- Alerts for unusual activity

3.2 User Creation & Management (Core Module)

Handles user lifecycle.

📌 **User Types:**

- Students
- Faculty
- Staff
- Parents
- Admins

⚙️ **Features:**

- Create users
- Edit profiles
- Activate/deactivate users
- Bulk import users

3.3 Role Management

Defines user roles.

⚙️ **Features:**

- Create roles (Principal, Teacher, Accountant, etc.)
- Modify roles
- Assign roles to users

3.4 Permission Management (RBAC Engine)

Controls what users can access.

⚙️ **Features:**

- Assign permissions per workspace
- Define access levels (view/edit/approve)
- Restrict sensitive data

3.5 Workspace Access Control

Maps users to workspaces.

⚙️ **Features:**

- Assign workspace access
- Control module-level permissions
- Enable/disable features

3.6 Authentication Management

Handles login security.

⚙️ **Features:**

- Username/password login
- OTP-based login (optional)
- Multi-factor authentication (MFA)

3.7 User Profiles & Identity (VID Profiles)

Maintains unified identity.

⚙️ **Features:**

- Unique VID ID per user
- Profile management
- Linked accounts (parent-student)

3.8 Audit Logs & Activity Tracking

Tracks user activity.

⚙️ **Features:**

- Login logs
- Action logs
- Security monitoring

3.9 Session Management

Controls user sessions.

⚙️ **Features:**

- Active session tracking
- Force logout
- Session expiry control

3.10 Bulk Operations

Handles large-scale user operations.

⚙️ **Features:**

- Bulk user upload
- Bulk role assignment
- Bulk updates

3.11 Notifications & Alerts

Security-related alerts.

⚙️ **Alerts:**

- Failed login attempts
- Suspicious activity
- Password changes

3.12 Reports & Analytics

Provides user insights.

📊 **Reports:**

- User activity report
- Role usage report
- Login trends

3.13 Data Privacy & Compliance

Ensures legal compliance.

⚙️ **Features:**

- Data access restrictions
- Privacy controls
- Compliance logs

3.14 Configuration Module

Controls UMS settings.

⚙️ **Features:**

- Password policies
- Login rules
- Role defaults

3.15 "What's New" Module

Displays updates/features.

4. 🔄 End-to-End Workflow

1. Create user
2. Assign role
3. Define permissions
4. Grant workspace access
5. User logs in
6. System validates access
7. Track activity

5. 🧠 Data Ownership

This workspace manages:

- User profiles
- Roles & permissions
- Authentication data
- Activity logs

6. 🔗 Dependencies

Controls Access To:
👉 All Workspaces

Receives Data From:

- Team Owner Workspace (global control)

7. 📊 Key Metrics

- Active users
- Login success rate
- Security incidents
- Role distribution

8. 🖥️ UI / UX Structure (Recommended)

Sidebar:

- Dashboard
- Users
- Roles
- Permissions
- Sessions
- Reports
- Configuration

Main UI:

User Table:

- Name | Role | Status

Permission Matrix:

- Role vs Workspace Access

9. ⚠️ Critical Design Notes

- Must enforce **strict RBAC**
- Needs **high security standards**
- Should support **scalability (thousands of users)**
- Must ensure **data privacy compliance**

10. 🚀 Why This Workspace is Critical

Without this:

- No access control
- No security
- No role-based system

👉 This workspace is the **security and identity backbone of the entire VID platform**

AI AGENT 1: CHATBOT (YANTRA)

1. 🧭 Overview

The **Chatbot Agent** is an intelligent, conversational AI system embedded across the VID platform to provide **real-time assistance, data access, and workflow support**.

It is not just a FAQ bot — it is:
👉 A **data-aware assistant**
👉 A **system navigator**
👉 A **decision support interface**

It connects with:

- Student Workspace
- Faculty Workspace
- Payments
- Examination
- HRMS
- Common Workspace

2. 🎯 Core Purpose

The chatbot is designed to:

- Reduce manual dependency on staff
- Provide instant answers to users
- Improve platform usability
- Automate repetitive queries

3. 👥 User Coverage

The chatbot is **multi-role adaptive**:

🎓 **Students**

- Attendance queries
- Exam schedules
- Homework reminders

👩‍🏫 **Faculty**

- Class data
- Student performance
- Timetable access

👪 **Parents**

- Fee status
- Attendance alerts
- School updates

🏢 **Admin**

- Reports

- Analytics queries
- Quick system access

4. 🧩 Core Features (Detailed)

4.1 Intelligent Query Resolution (Core Engine)

This is the **heart of the chatbot**.

🧠 **Capabilities:**

- Understand natural language
- Context-aware responses
- Multi-step query handling

💬 **Examples:**

- "Show my attendance this month"
- "When is my next exam?"
- "How much fee is pending?"

⚙️ **Backend Behavior:**

- Detect intent
- Identify user role
- Fetch relevant data
- Format response

4.2 Context-Aware Personalization

The chatbot adapts based on **who is asking**.

🎯 **Example:**

Same question: "Show attendance"

- Student → Shows personal attendance
- Parent → Shows child's attendance
- Admin → Shows class/institution summary

4.3 Deep Workspace Integration

The chatbot pulls real-time data from all workspaces.

🔗 **Integrations:**

- **Student Workspace** → Attendance, marks
- **Payment Workspace** → Fees, dues
- **Examination Workspace** → Results
- **Timetable Workspace** → Schedules
- **HRMS** → Staff data

4.4 Smart Navigation Assistant

Acts as a **UI shortcut system**.

⚙️ **Features:**

- Opens modules directly
- Redirects users

💬 **Examples:**

- "Open timetable"
- "Go to payments"

👉 Instantly navigates to that page

4.5 Proactive Notifications & Reminders

Not just reactive — also proactive.

🔔 **Alerts:**

- Fee due reminders
- Exam schedules
- Event alerts
- Assignment deadlines

4.6 Knowledge Base & FAQ Engine

Pre-trained institutional knowledge.

📚 **Includes:**

- School policies
- Admission procedures
- Rules & regulations

4.7 Multi-Language Support (Critical for India)

🌐 **Supports:**

- English
- Regional languages (Telugu, Hindi, etc.)

👉 Expands accessibility across users

4.8 Conversation Memory (Context Retention)

Remembers previous interactions.

🧠 **Example:**

User: "Show attendance"
Then: "What about last month?"

👉 It understands continuity

4.9 Escalation System (Human Support)

If AI cannot answer:

⚙️ **Features:**

- Escalates to admin/support

- Creates support tickets

4.10 Chat Analytics & Insights

Tracks chatbot usage.

📊 **Metrics:**

- Queries handled
- Success rate
- Common questions

5. 🔄 End-to-End Workflow

User asks question →

AI processes intent →

Identifies role →

Fetches data from workspace →

Formats response →

Returns answer →

Suggests next actions

6. 🧠 Data Access Layer

The chatbot is a **read-heavy system**:

- Student data
- Attendance
- Fees
- Exams
- Timetable

👉 Controlled via RBAC (UMS)

7. 🖥️ UI/UX Placement

📍 **Entry Points:**

- Floating chat icon (bottom-right)
- Dashboard widget
- Mobile chat interface

💬 **Chat Window Design:**

- Input box
- Suggested queries
- Quick action buttons

8. ⚠️ Design Considerations

- Must be **fast (<2 sec response)**
- Needs **high accuracy**

- Should handle **ambiguous queries**
- Must respect **data privacy (RBAC)**

9. 🚀 Strategic Importance

This is one of your **biggest differentiators**.

👉 Reduces workload on staff
👉 Improves user experience
👉 Makes system intelligent

🔥 **Final Positioning**

👉 Chatbot = **Frontline AI Interface of VID**

🟦 AI AGENT 2: VOICE AGENT (YANTRA)

1. 🧭 Overview

The **Voice Agent** is an AI-powered **speech-based interaction system** that allows users to operate the VID platform using **voice commands instead of manual navigation**.

It transforms the system into:
👉 A **hands-free operational platform**
👉 A **natural interaction interface**

It works as:

- A **voice-controlled assistant**
- A **command execution engine**
- A **spoken data retrieval system**

2. 🎯 Core Purpose

The Voice Agent is designed to:

- Reduce time spent navigating UI
- Enable faster execution of tasks
- Improve accessibility for all users
- Provide a modern AI-driven experience

3. 👥 User Coverage

👑 Admin / Principal

- Quick report access
- Monitoring operations
- Voice-based commands

👩‍🏫 Faculty

- Attendance marking
- Class navigation
- Data retrieval

👪 Parents

- Fee queries
- Attendance updates

👨‍💼 Staff

- Reports
- Requests
- Workflow execution

4. 🧩 Core Features (Detailed)

4.1 Voice Command Execution (Core Engine)

This is the **primary capability**.

🎙 **Examples:**

- “Show today’s attendance”
- “Open fee reports”
- “Create announcement”
- “Go to timetable”

⚙ **How it works:**

1. Speech → converted to text
2. AI detects intent
3. Maps to system action
4. Executes command

4.2 Voice-Based Navigation

Replaces manual clicking.

⚙ **Features:**

- Open workspaces
- Switch modules
- Navigate dashboards

🎯 **Example:**

“Open examination workspace”
👉 Direct navigation

4.3 Data Retrieval via Voice

Users can **ask and listen to data**.

🎙 **Examples:**

- “What is today’s attendance?”
- “How much fee is collected?”

⚙ **Output:**

- Display on screen
- Read aloud via voice

4.4 Voice-Based Announcements

Create announcements using voice.

⚙ **Flow:**

Speak → Convert → Publish

🎙 **Example:**

“Announce holiday tomorrow”

👉 System creates notice automatically

4.5 Multi-Language Voice Support

Critical for real-world adoption.

🌐 **Supports:**

- English
- Telugu
- Hindi
- Other regional languages

4.6 Smart Command Suggestions

System helps users.

⚙️ **Features:**

- Suggest commands
- Auto-complete intent

4.7 Voice Feedback System

System responds back.

🔊 **Features:**

- Confirms actions
- Reads results

Example:
👉 "Attendance marked successfully"

4.8 Accessibility Enablement

Major advantage.

♿ **Helps:**

- Non-technical users
- Elder staff
- Visually impaired users

4.9 Workspace Integration

Works across all modules.

🔗 **Integrations:**

- Academic → data fetch
- Payment → reports
- HRMS → attendance
- Examination → results

4.10 Noise Handling & Accuracy Layer

Handles real-world conditions.

⚙️ **Features:**

- Noise filtering
- Voice clarity processing
- Error correction

5. 🔄 End-to-End Workflow

User speaks →

Speech-to-text conversion →

AI processes intent →

Maps to system action →

Executes command →

Returns result (UI + voice)

6. 🧠 Data Access & Execution Layer

The Voice Agent:

- Reads data (like chatbot)
- Executes actions (unlike chatbot)

👉 Controlled via RBAC permissions

7. 🖥️ UI/UX Placement

📍 **Entry Points:**

- 🎙️ Mic button (Top Navbar)
- Floating voice button
- Dashboard widget

🎛️ **Voice Interface UI:**

- Mic ON/OFF indicator
- Listening animation
- Transcript display

8. ⚠️ Design Considerations

- Must be **highly accurate in speech recognition**
- Needs **fast processing (<2–3 seconds)**
- Should handle **multiple accents**
- Must ensure **secure command execution**

9. 📊 Key Metrics

- Voice command accuracy
- Execution success rate
- Average response time
- User adoption rate

10. 🚀 Strategic Importance

👉 Makes system **future-ready**
👉 Reduces dependency on UI complexity
👉 Improves speed of operations

🔥 **Final Positioning**

👉 Voice Agent = **Command Layer of VID (Action Engine)**

AI AGENT 3: FACE ATTENDANCE SYSTEM (YANTRA)

1. 🧭 Overview

The Face Attendance Agent is an AI-powered system that automates attendance using facial recognition technology.

It replaces:
❌ Manual roll calls
❌ ID cards / RFID systems

It enables:
👉 Fully automated attendance
👉 Real-time tracking
👉 High accuracy monitoring

It acts as:

- A biometric attendance system
- A real-time monitoring engine
- A discipline enforcement tool

2. 🎯 Core Purpose

The system is designed to:

- Eliminate manual attendance errors
- Save classroom time
- Prevent proxy attendance
- Provide real-time attendance insights

3. 👥 User Coverage

🎓 Students

- Automated attendance marking

👩‍🏫 Faculty

- Monitor attendance
- Validate records

🏢 Admin

- View analytics
- Track discipline

👪 Parents

- Get attendance alerts

4. 🧩 Core Features (Detailed)

4.1 Face Recognition Engine (Core AI)

This is the foundation of the system.

⚙️ Capabilities:

- Capture face via camera
- Detect facial features
- Match with stored database

🧠 Process:

- Face detection
- Feature extraction
- Face embedding creation
- Matching with stored profiles

4.2 Automated Attendance Marking

Marks attendance instantly.

⚙️ Features:

- Classroom attendance
- Entry/exit tracking
- Period-wise attendance

📌 Example:

Student enters classroom → camera detects → attendance marked

4.3 Real-Time Camera Integration

Works with live video feeds.

⚙️ Sources:

- CCTV cameras
- Device cameras (mobile/web)

4.4 Attendance Logs & Reports

Maintains complete history.

📊 Includes:

- Daily attendance
- Monthly summaries
- Absentee reports
- Late entry tracking

4.5 Anti-Spoofing & Liveness Detection

Prevents fraud.

⚙️ Features:

- Detect printed photos
- Detect screen replays
- Ensure real human presence

4.6 Multi-Face Detection

Handles groups.

⚙️ Features:

- Detect multiple students at once
- Batch attendance marking

4.7 Alerts & Notifications

Real-time updates.

🔔 Alerts:

- Absent notification to parents
- Late arrival alerts
- Irregular attendance alerts

4.8 Integration with Timetable

Smart mapping.

⚙️ Features:

- Links attendance to class schedule
- Period-based validation

4.9 Integration with Workspaces

🔗 Connected to:

- Student Workspace → attendance records
- Faculty Workspace → monitoring
- HRMS → staff attendance
- Timetable Workspace → scheduling logic

4.10 Attendance Correction System

Handles errors.

⚙️ Features:

- Manual override
- Correction requests
- Approval workflow

4.11 Analytics & Insights

Provides insights.

📊 Metrics:

- Attendance percentage
- Class-wise trends
- Student discipline patterns

4.12 Offline Mode (Optional but Powerful)

Works without internet.

⚙️ Features:

- Local data capture
- Sync when online

5. 🔄 End-to-End Workflow

Camera captures face →

AI detects face →

Matches with database →

Identifies user →

Marks attendance →

Updates system →

Triggers alerts (if needed)

6. 🧠 Data Ownership

This system manages:

- Face embeddings (biometric data)
- Attendance logs
- Recognition records

👉 Highly sensitive → must be secured

7. 🖥️ UI/UX Placement

📍 Interfaces:

- Attendance dashboard
- Camera screen
- Reports panel

📹 Camera UI:

- Live feed
- Face detection boxes
- Status (recognized / not recognized)

8. ⚠️ Critical Design Considerations

🔐 Privacy & Security

- Encrypt biometric data
- Follow compliance standards
- Restricted access via RBAC

🎯 Accuracy Requirements

- Must work in:

    - Low light
    - Different angles
    - Large classrooms

⚡ Performance

- Real-time processing
- Minimal delay

🧬 Edge Case Handling

- Identical twins
- Masked faces
- Partial visibility

9. 📊 Key Metrics

- Recognition accuracy (%)
- False positives/negatives
- Attendance automation rate
- Processing time

10. 🚀 Strategic Importance

This is a high-impact innovation feature.

👉 Saves time daily
👉 Improves discipline
👉 Eliminates proxy attendance
👉 Enhances trust with parents

🔥 Final Positioning

👉 Face Attendance = Automation Engine of VID